M.TECH THESIS PROGRESS

LES MODELING OF TURBULENT FLOW

By : Aditya Jain
23110017

Key Learnings so far :

tutorials/incompressible/icoFoam/cavity/cavity

- Tutorials
  - Example cases provided by OpenFOAM
- incompressible
  - Constant Density
- icoFoam
  - Solver for laminar incompressible flow
- cavity
- cavity

Cavity(These are Folders)

- 0/
  Determines flow start
- Constant
  Details of Fluid and mesh
- System
  Numerical Solver

controlDict :

1. Which solver to run?
2. When to start the simulation?
3. When to stop?
4. How big is each time step?
5. How often should the results be saved?

Boundary Names:

- patch 0 name: movingWall
- patch 1 name: fixedWalls
- patch 2 name: frontAndBack

Dimensions vector:
[mass length time temperature current amount intensity]

Simulation 1: Lid Driven Cavity

Problem Statement:

- A square box filled with fluid
- All four walls are solid
- The top wall moves sideways at 1 m/s
- Other three walls are stationary
- Fluid starts at rest
- Motion spreads inward due to viscosity

Given:

- Square box : 1m X 1m
- Ux = 1 m/s
- Re = 100
- No slip at walls

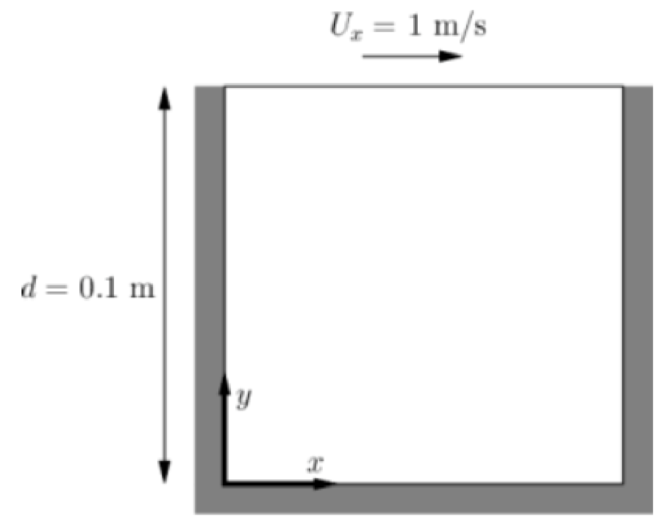


Figure 2.1: Geometry of the lid driven cavity.

Solver Properties

Solver Used : IcoFoam

- Designed for **laminar, incompressible flow of Newtonian fluids**
- Uses PISO algorithm, it is inherently designed for transient simulations
- need to define one physical constant: the kinematic viscosity

Equations

The solver uses the **PISO** algorithm to solve the continuity equation:

$$\nabla \cdot \mathbf{u} = 0$$

and momentum equation:

$$\frac{\partial}{\partial t}(\mathbf{u}) + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) - \nabla \cdot (\nu \nabla \mathbf{u}) = -\nabla p$$

Where:

$\mathbf{u}$ = Velocity

p = Kinematic pressure

Simulation 1: Lid Driven Cavity

Velocity Boundary Conditions:

Patch	Type	Values	Physical meaning
movingWall	fixedValue	(1 0 0)	Wall moves at 1 m/s in the x-direction. Drives the lid-driven cavity flow.
fixedWalls	noSlip	(0 0 0)	No-slip condition. Fluid velocity at the wall equals zero (walls are stationary).
frontAndBack	empty	—	2-D simulation plane. OpenFOAM applies no solution in this direction.

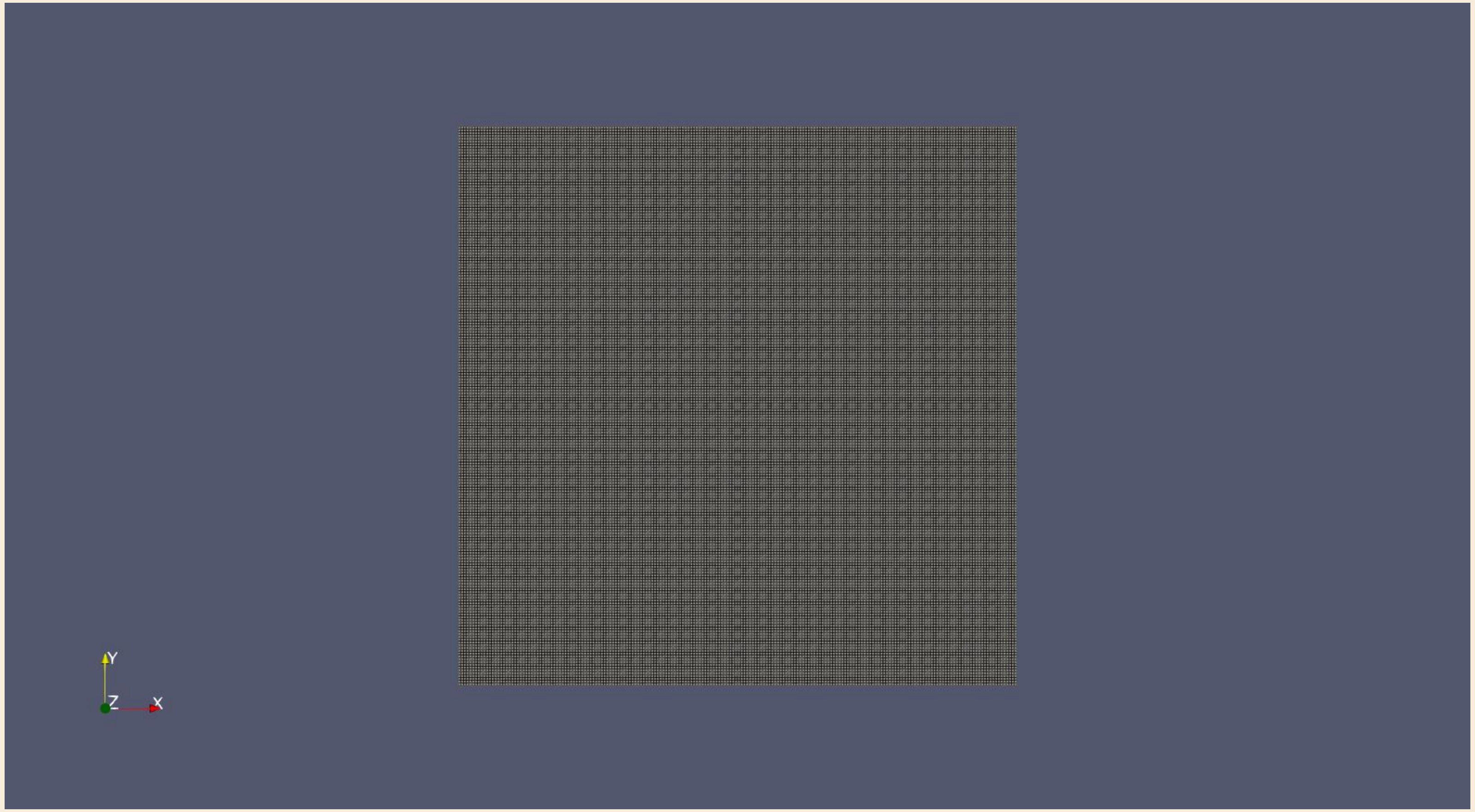
Pressure Boundary Conditions:

Patch	Type	BC class	Physical meaning
movingWall	zeroGradient	fixedGradient	No normal pressure gradient across the wall. Pressure value extrapolated from interior.
fixedWalls	zeroGradient	fixedGradient	No normal pressure gradient across the wall. Pressure value extrapolated from interior.
frontAndBack	empty	empty	2-D simulation plane. OpenFOAM applies no solution in this direction.

Mesh Details

Mesh Wireframe

Property	Value
Domain	1 m × 1 m
Mesh type	Structured hexahedral
Grid resolution	200 × 200
Total cells	40,000
Cell size (Δx = Δy)	0.005 m
Grading	Uniform (1 1 1)



Velocity Plot

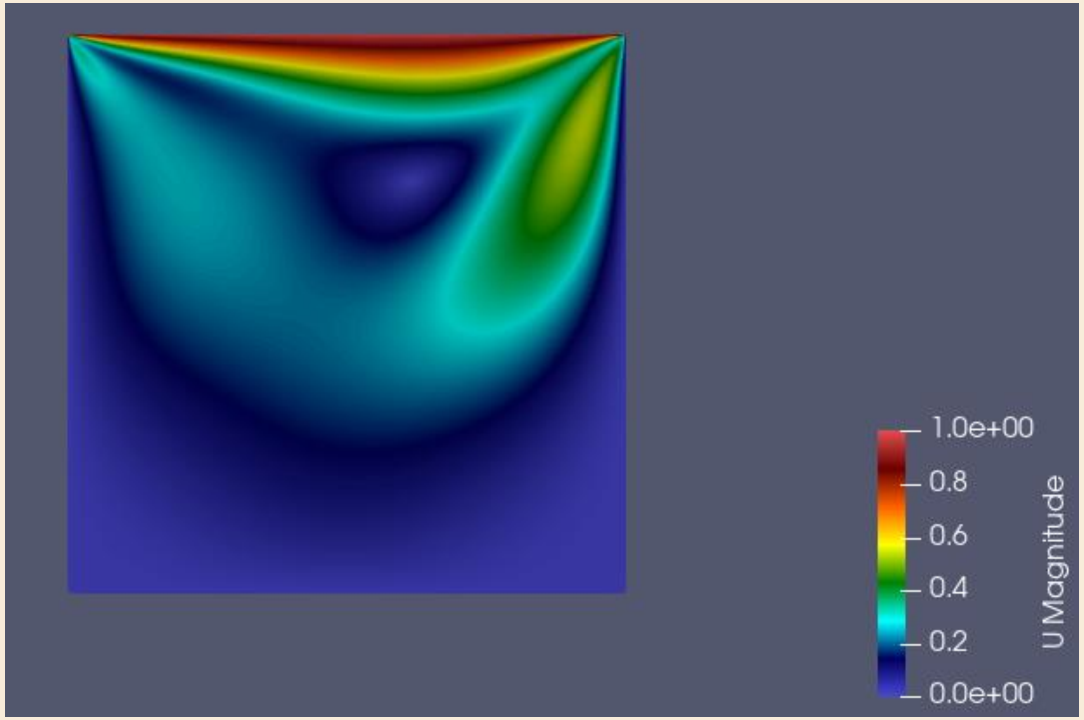
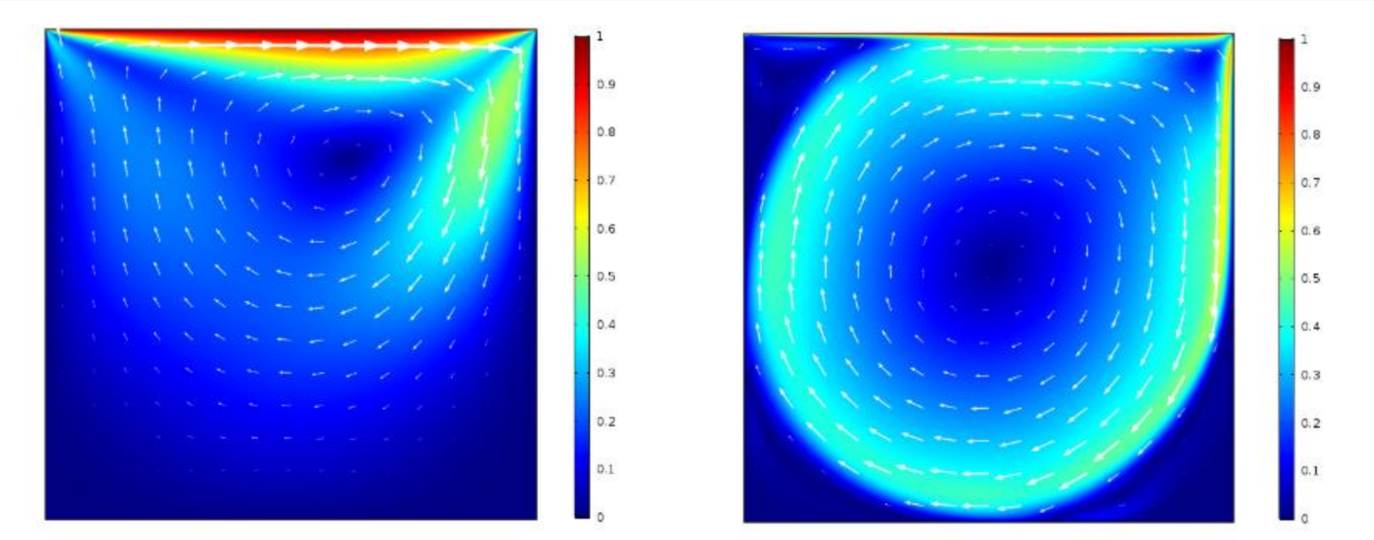


Fig. OpenFoam Velocity Contour plot, Re=100

Fig. Plot for Validation, Re=100

https://www.comsol.com/blogs/how-to-solve-a-classic-cfd-benchmark-the-lid-driven-cavity-problem



The magnitude of the velocity and the direction of the flow in the cavity for Reynolds numbers of 100 (left) and 10,000 (right).

Cavity flow validation

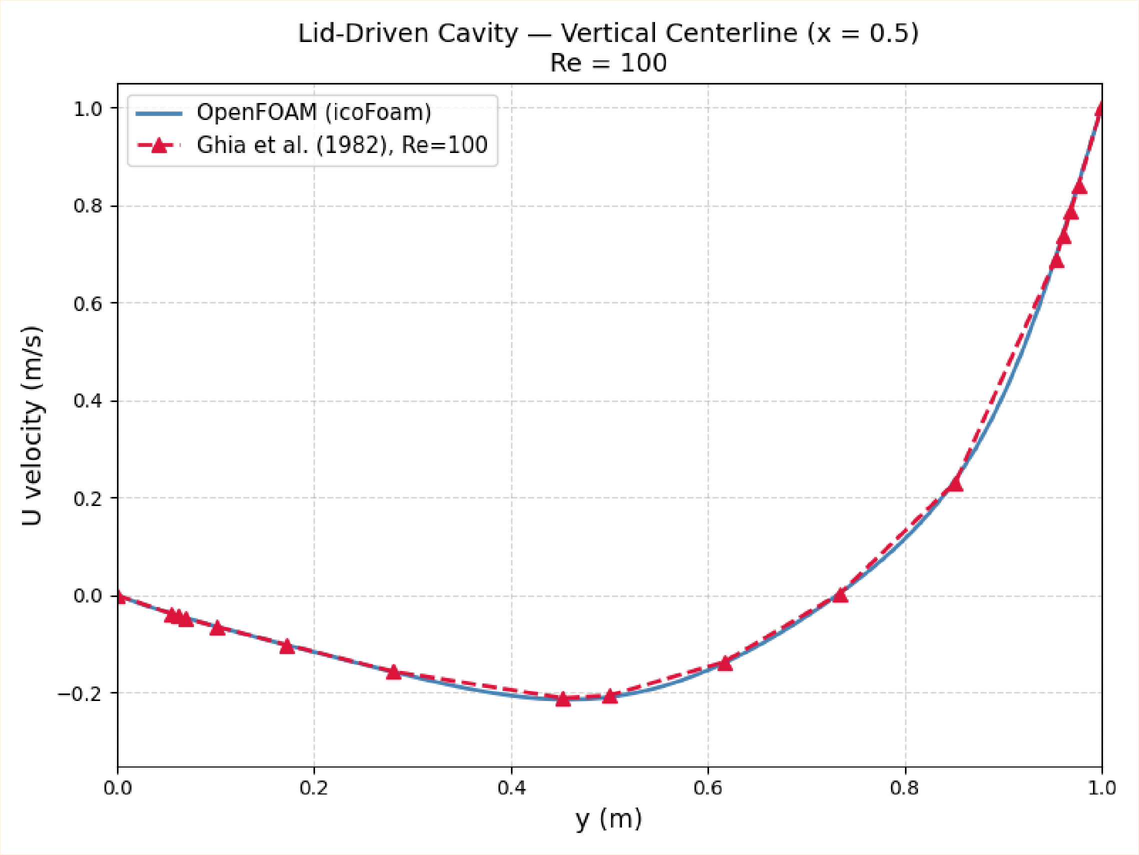


Fig. Comparison of the computed horizontal velocity (u) along the vertical centerline against benchmark data.

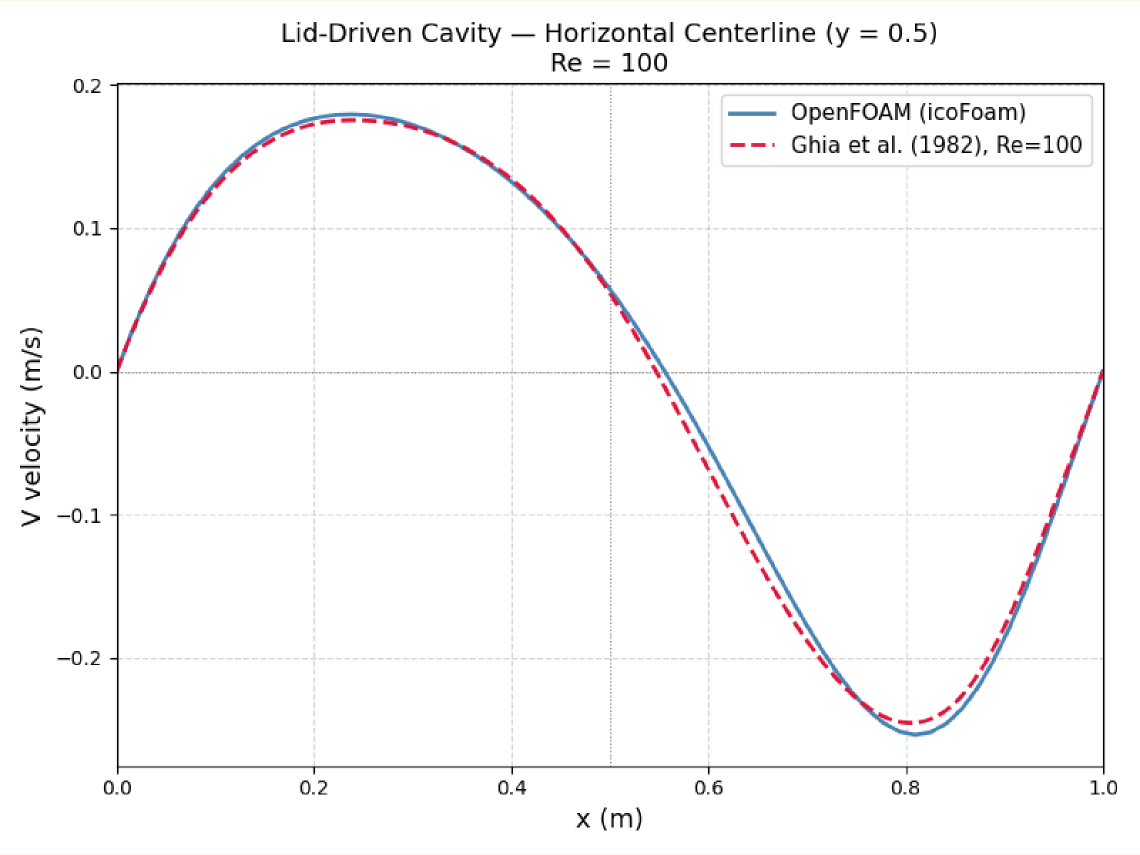


Fig. Comparison of the computed vertical velocity (v) along the horizontal centerline against benchmark data.

Simulation 2: Flow Over a Cylinder

Problem Statement:

- The problem statement for flow over a cylinder involves simulating a **viscous fluid** moving past a fixed circular obstacle to analyze how the presence of the body alters the flow field. Specifically, it aims to calculate the **velocity and pressure distributions** to determine the transition from steady flow to periodic **vortex shedding** based on the Reynolds number.

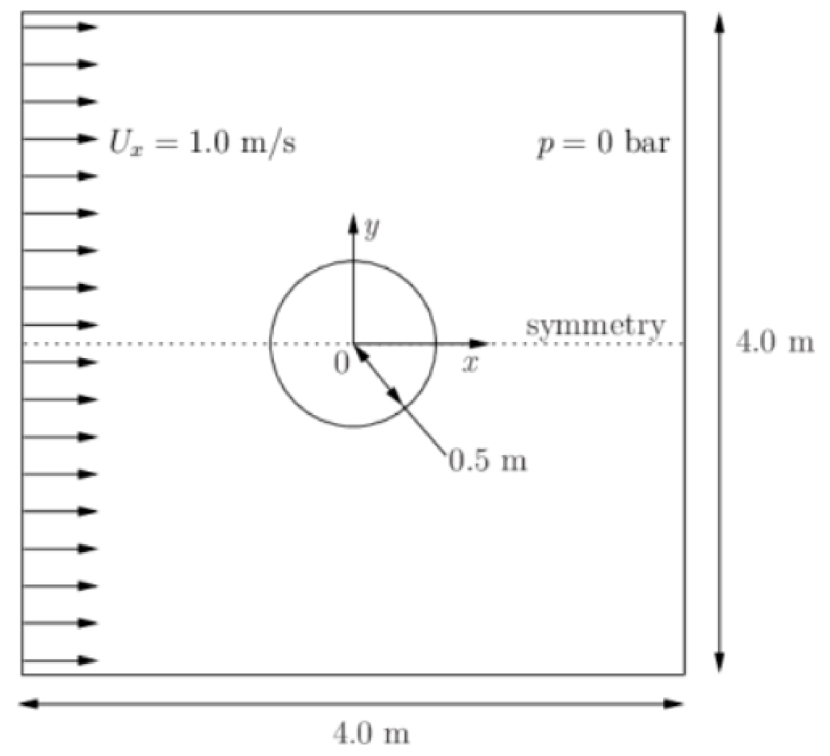
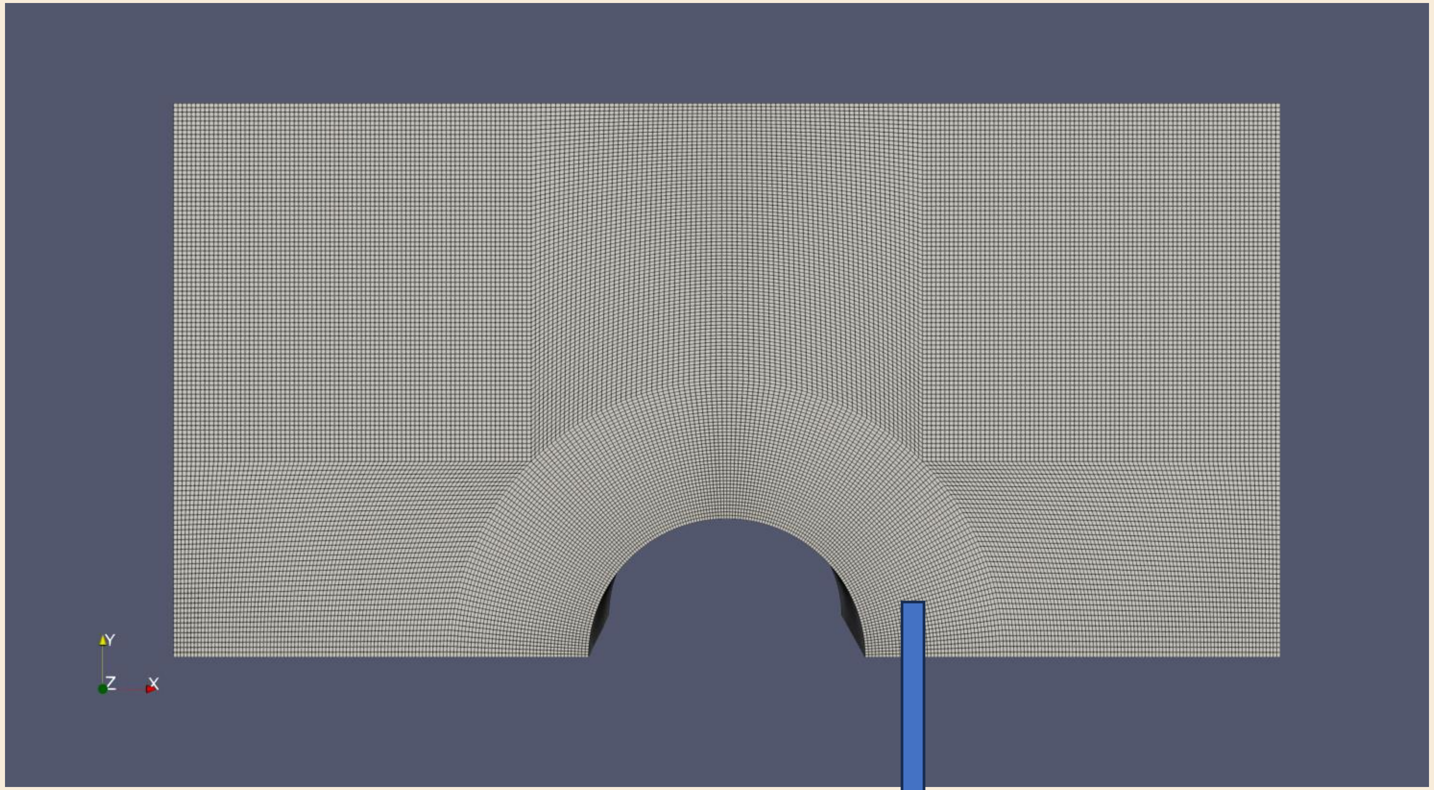
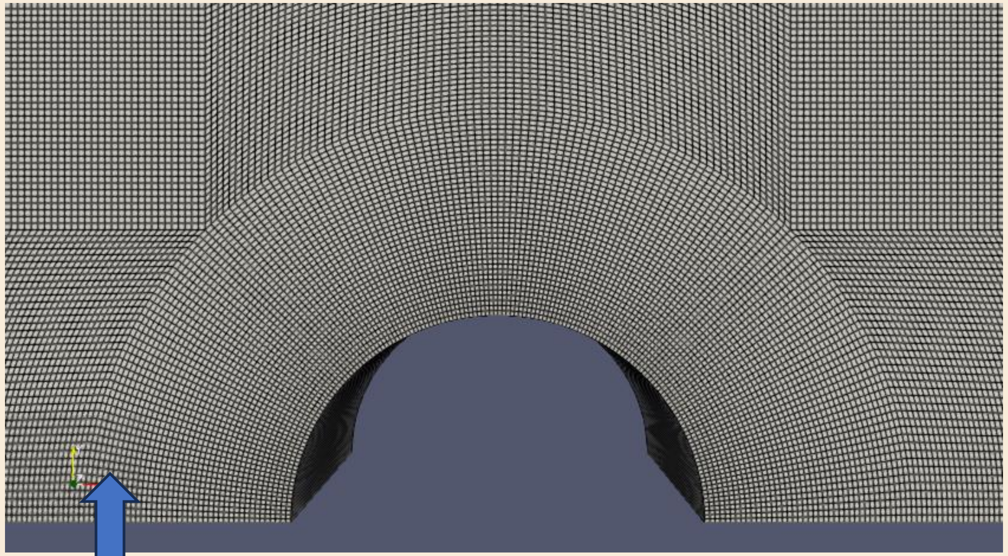


Figure 2.15: Geometry of flow round a cylinder

Mesh Generation





Mesh Information

Points: 16442

Cells: 8000

Faces: 32220

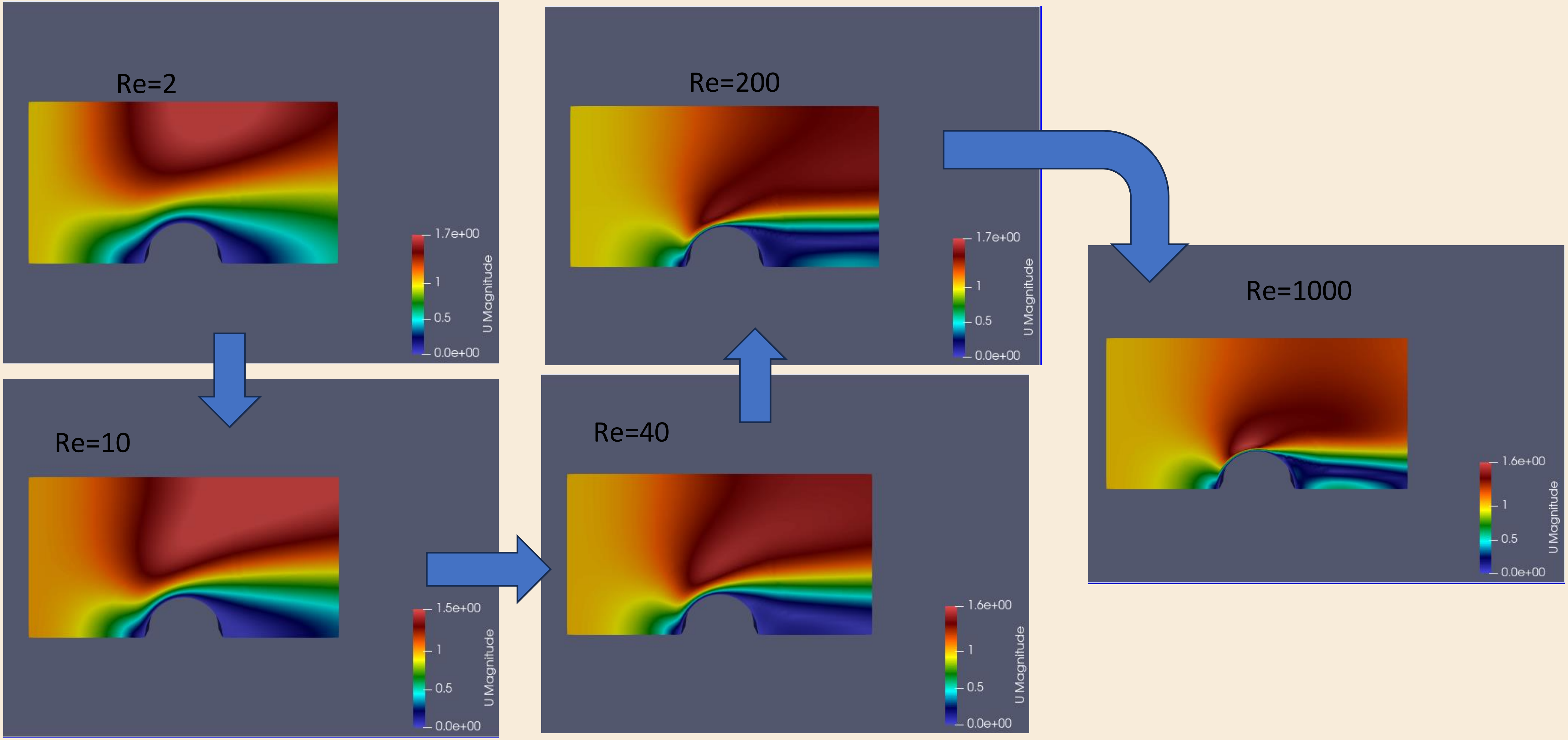
Internal Faces: 15780

potentialFoam VS icoFoam

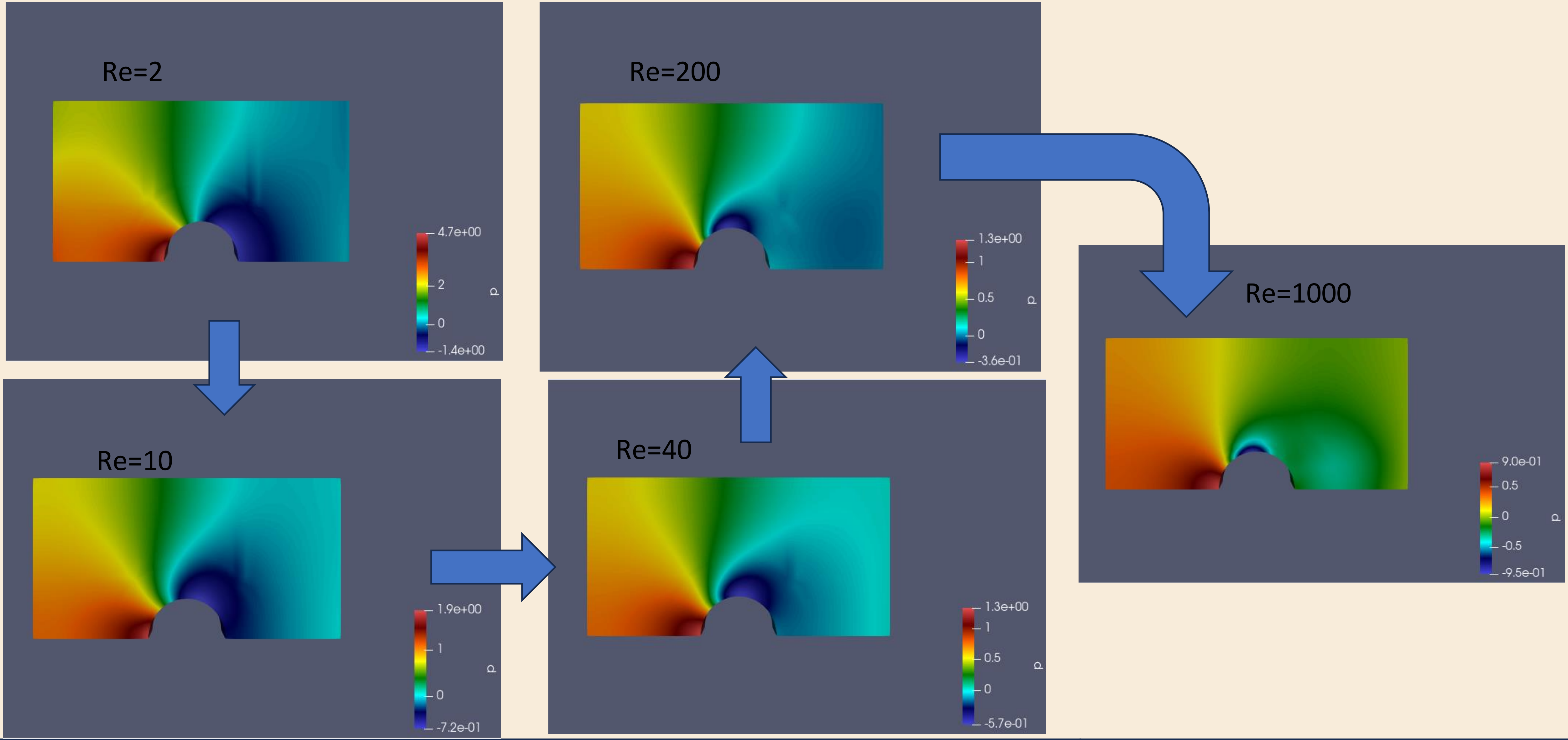
Aspect	potentialFoam	icoFoam
Type of solver	Steady	Transient (time-dependent)
Physics	Inviscid, irrotational	Viscous, solves full Navier–Stokes
Viscosity (ν)	Not required	Must define in transportProperties
Time stepping	Single step	Many time steps
Required files	controlDict only	controlDict + fvSchemes + fvSolution + transportProperties
Numerical schemes	Very minimal	Needs ddtSchemes, divSchemes, laplacianSchemes
Pressure–velocity coupling	Not required	Uses PISO algorithm
Cylinder wall BC	Often symmetry	Must be noSlip
Flow features	No boundary layer	Boundary layer forms
Validation possible with theory	Yes (analytical solution exists)	No (viscous effects present)

Therefore, **icoFoam** was used for simulation for flow over a cylinder.

Results: Velocity Magnitude

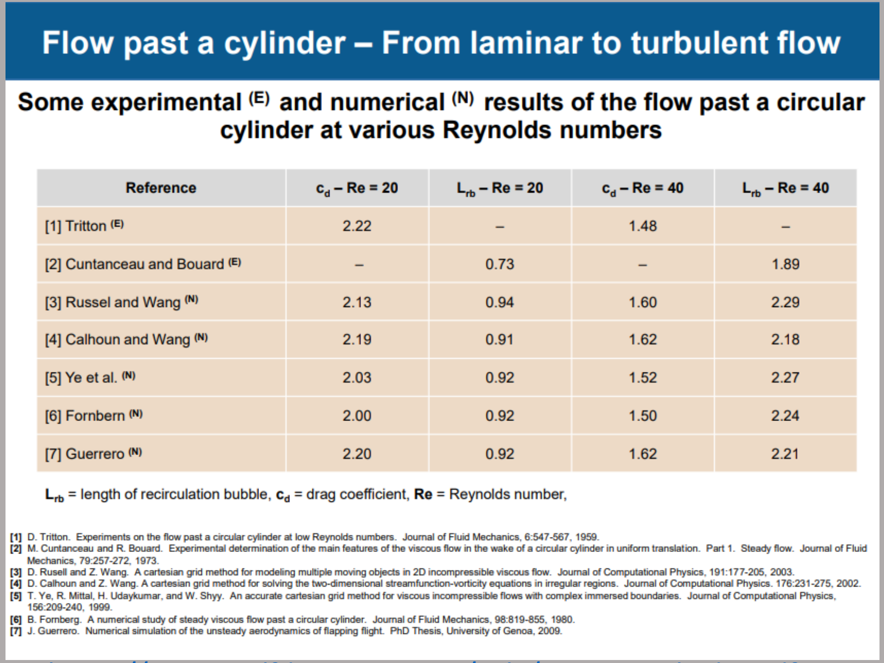


Results: Pressure plot



Verification : Drag Coefficient Comparison

Fig. Reference Results to compare with



Flow past a cylinder – From laminar to turbulent flow

Some experimental (E) and numerical (N) results of the flow past a circular cylinder at various Reynolds numbers

Reference	c_d – Re = 20	L_{rb} – Re = 20	c_d – Re = 40	L_{rb} – Re = 40
[1] Tritton (E)	2.22	–	1.48	–
[2] Cuntanceau and Bouard (E)	–	0.73	–	1.89
[3] Russel and Wang (N)	2.13	0.94	1.60	2.29
[4] Calhoun and Wang (N)	2.19	0.91	1.62	2.18
[5] Ye et al. (N)	2.03	0.92	1.52	2.27
[6] Fornbern (N)	2.00	0.92	1.50	2.24
[7] Guerrero (N)	2.20	0.92	1.62	2.21

L_{rb} = length of recirculation bubble, c_d = drag coefficient, **Re** = Reynolds number,

[1] D. Tritton. Experiments on the flow past a circular cylinder at low Reynolds numbers. Journal of Fluid Mechanics, 6:547-567, 1959.
[2] M. Cuntanceau and R. Bouard. Experimental determination of the main features of the viscous flow in the wake of a circular cylinder in uniform translation. Part 1. Steady flow. Journal of Fluid Mechanics, 79:257-272, 1973.
[3] D. Rusell and Z. Wang. A cartesian grid method for modeling multiple moving objects in 2D incompressible viscous flow. Journal of Computational Physics, 191:177-205, 2003.
[4] D. Calhoun and Z. Wang. A cartesian grid method for solving the two-dimensional streamfunction-vorticity equations in irregular regions. Journal of Computational Physics. 176:231-275, 2002.
[5] T. Ye, R. Mittal, H. Udaykumar, and W. Shyy. An accurate cartesian grid method for viscous incompressible flows with complex immersed boundaries. Journal of Computational Physics, 156:209-240, 1999.
[6] B. Fornberg. A numerical study of steady viscous flow past a circular cylinder. Journal of Fluid Mechanics, 98:819-855, 1980.
[7] J. Guerrero. Numerical simulation of the unsteady aerodynamics of flapping flight. PhD Thesis, University of Genoa, 2009.

https://www.wolfdynamics.com/wiki/tut_2D_cylinder.pdf

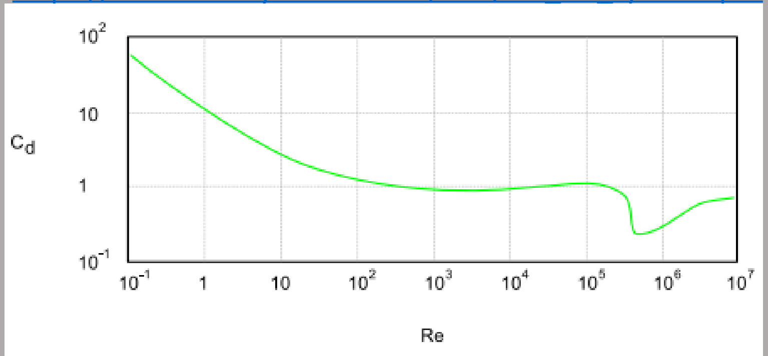
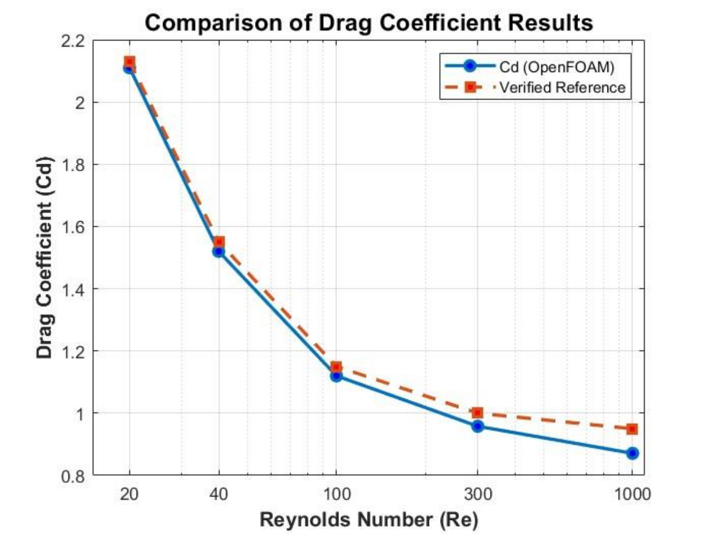


Fig. OpenFOAM results

Re	Cd(OpenFOAM)	Verified reference values
20	2.11	2.13
40	1.52	1.55
100	1.12	1.15
300	0.958	1
1000	0.871	0.95



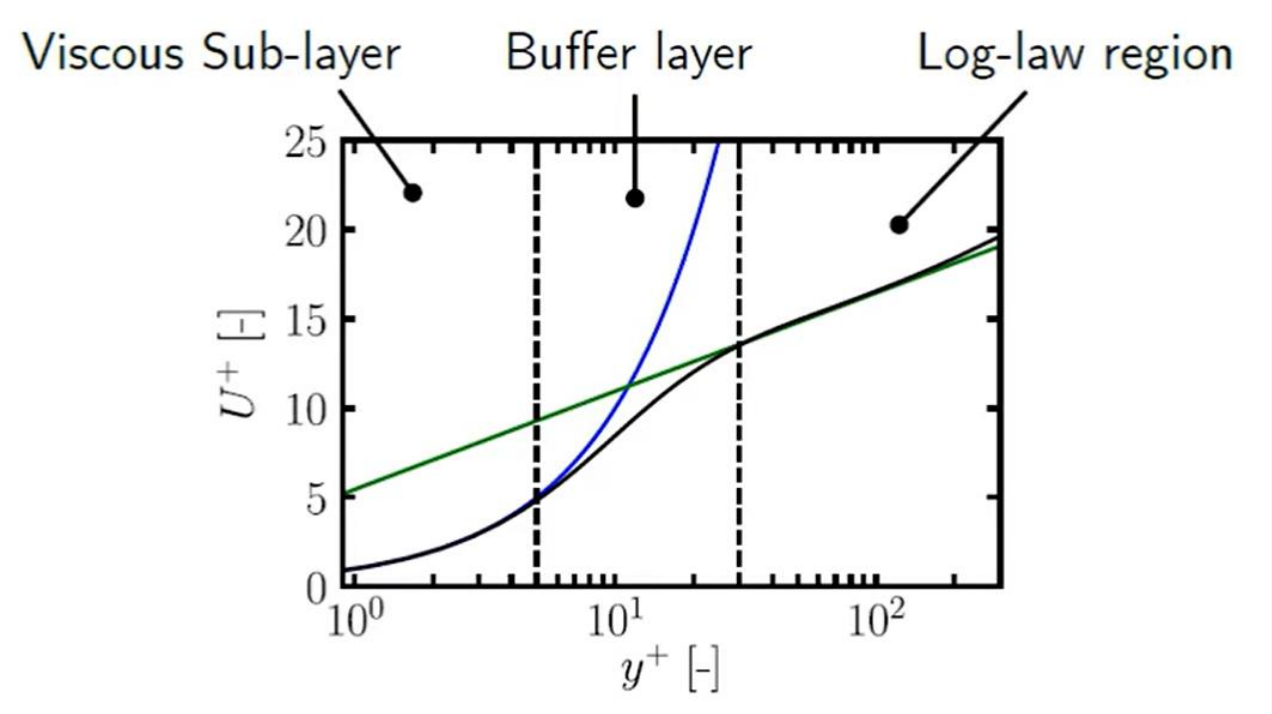
Understanding Turbulence Modelling

Velocity Distribution in a Turbulent Boundary Layer

- Near the wall (Viscous Sublayer, $y^+ < 5$) :Flow is dominated by molecular viscosity and the velocity increases linearly with distance from the wall $(U^+ = y^+)$.

- Intermediate region (Buffer Layer, $5 < y^+ < 30$:(This is a transition zone where both viscous and turbulent stresses influence the flow, leading to a gradual shift in the velocity profile.

- Outer region (Log-law Region, $y^+ > 30$:(Turbulent mixing dominates, and the velocity follows the logarithmic law of the wall, increasing logarithmically with distance from the wall.



Region	Range of y^+	Dominant Physics	Velocity Relation
Viscous Sublayer	$y^+ < 5$	Molecular viscosity dominates	$U^+ = y^+$
Buffer Layer	$5 < y^+ < 30$	Transition between viscous and turbulent stresses	No simple analytical relation
Log-Law Region	$y^+ > 30$	Turbulent mixing dominates	$U^+ = \frac{1}{\kappa}\ln(Ey^+)$

Empirical coefficients:
- κ = 0.4187
- E = 9.793

k-ϵ Turbulence Model

- RANS equation :

$$\frac{d\rho U}{dt} + \nabla.(\rho UU) = -\nabla p + \nabla[\mu(\nabla U + (\nabla U)^T] + \rho g - \nabla\left(\frac{2}{3}\mu(\nabla.U)\right) - \nabla.(\overline{\rho U'U'})$$

- The Boussinesq hypothesis: proposes that the momentum transfer caused by turbulent eddies can be modeled using an **eddy viscosity (μ_t),** similar to molecular viscosity.

- Scale of Turbulence :

$$\mu_t = \rho k^{1/2} l_m$$

where l_m is mixing length, this is an algebraic relation.

Thus we solve transport equation (a fundamental linear partial differential equation (PDE) that models the movement of a scalar) for k and ϵ, we get :

$$\mu_t = C_\mu \rho k^2/\epsilon$$

where ϵ : turbulence dissipation rate, k : turbulent kinetic energy, $C_\mu = 0.09$

k-ϵ Turbulence Model

- **Turbulent Reynolds number** is used in **damping functions** near the wall for mixing length.
- Re_T is the Reynolds number that characterises the strength of the near wall turbulence relative to viscosity.

$$Re_T = \frac{\rho k^2}{\mu \epsilon}$$

- Re_T is small when in viscous sub-layer, as viscous effects dominate.
- $k - \epsilon$ model is usually preferred for high Re applications (y+ >30).
- The standard $k{-}\epsilon$ model assumes fully turbulent flow and therefore cannot capture the viscosity-dominated physics of the viscous sublayer, so it relies on wall functions instead of resolving it.
- $k - \omega$ SST model is preferred for near wall simulation for accuracy.

Turbulence Intensity

- Turbulence intensity (I) is a dimensionless metric quantifying the strength of velocity fluctuations relative to the mean flow velocity ($I = \frac{u'}{U}$), usually expressed as a percentage.
- We require root mean square of the fluctuations for finding Turbulence Intensity.
- $u_{rms} = \sqrt{\frac{1}{3}(\overline{u'^2_x} + \overline{u'^2_y} + \overline{u'^2_z})}$
- $I = \frac{u_{rms}}{|U|}$

Category	Typical Range	Applications
Low	0%-1%	Towing tanks, wind tunnels
Medium	1%-5%	Pipes, ducting, airplanes
High	>5%	Turbomachinery, coastal flows

- Now, RANS models resolve turbulent kinetic energy, and it is hard to interpret values of tke, if the value is large or small, how it changes with change in inlet velocity etc. Thus we use Turbulence intensity.

Turbulence Intensity

$$k = \frac{1}{2}\left(\overline{u'^2_x} + \overline{u'^2_y} + \overline{u'^2_z}\right)$$

- Therefore

$$u_{rms} = \sqrt{\frac{2k}{3}}$$

- Finding Intensity:

$$I = \sqrt{\frac{2k}{3|U|^2}}$$

- Instead of plotting k, we can plot I which is easier to interpret, and **we use I to specify inlet boundary conditions for k**.

$$k = \frac{3}{2} I^2 |U|^2$$

Channel Flow

Problem Statement

•**Channel flow** is fluid flow through a **duct with four equal flat walls**, driven mainly by a pressure gradient.

•Due to the **no-slip condition**, velocity is **zero at all walls** and **maximum near the center** of the channel.

•**Viscous boundary layers develop along each wall**, influencing the overall velocity distribution.

•The velocity profile is **not perfectly parabolic** because the **four corners modify the flow structure**.

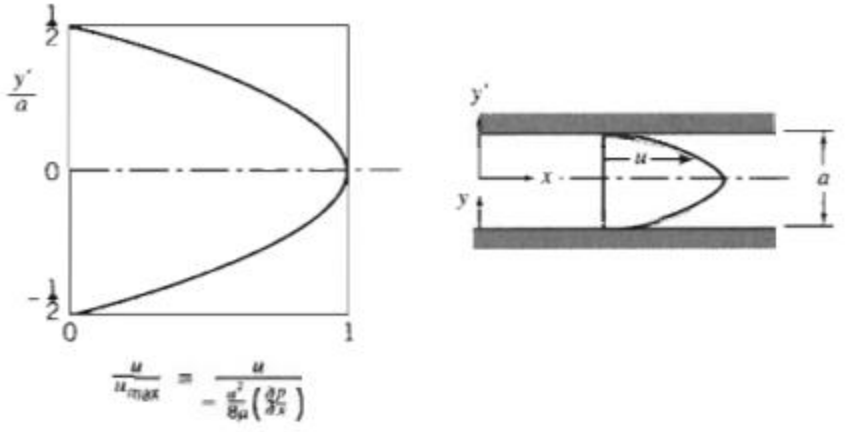
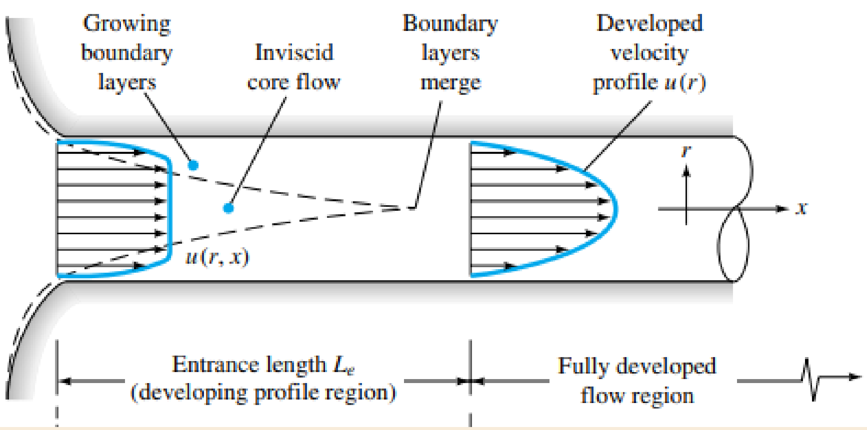


Fig. 8.3 Dimensionless velocity profile for fully developed laminar flow between infinite parallel plates.



k-ϵ solver for Channel Flow

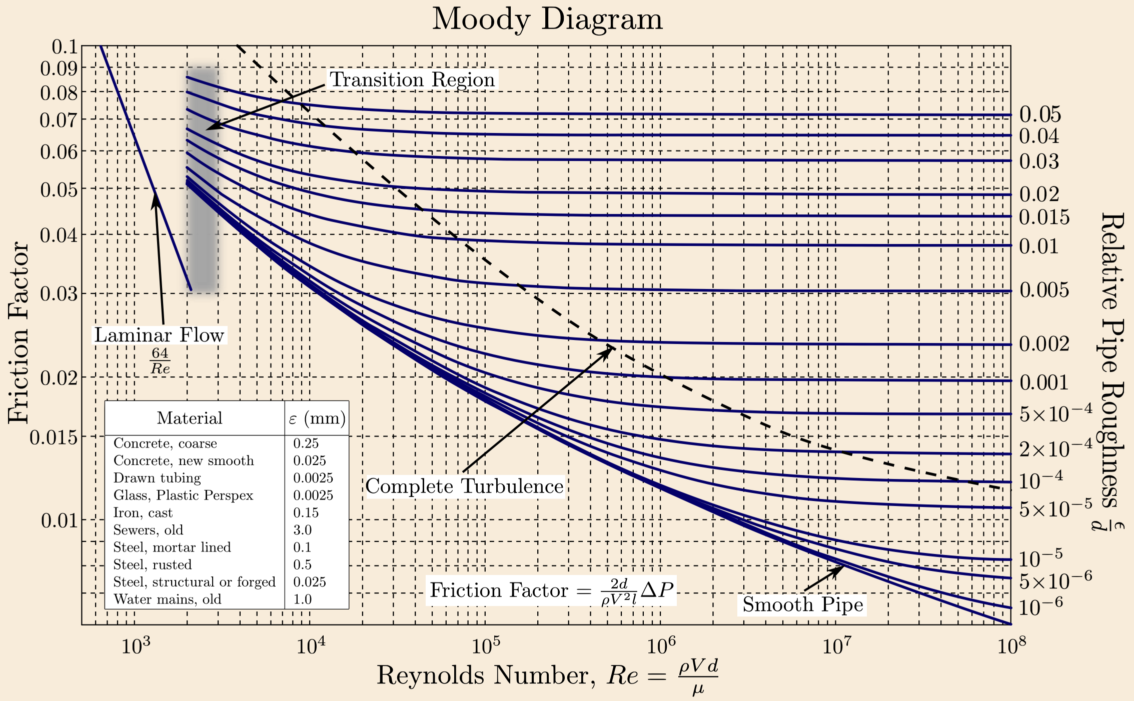
From Moody Diagram, at Re=66,667, $f_D \approx 0.017$

Therefore, $C_f = \frac{f_D}{4} = 0.00425$

$$Wall\ shear\ stress:\ \tau_w = C_f . \frac{1}{2} . \rho . U^2 = 0.24$$

$$Friction\ velocity:\ u_\tau = \sqrt{\frac{\tau_w}{\rho}} = 0.476\ \mathrm{m/s}$$

$$Viscous\ Length\ scale:\ \delta_\vartheta = \frac{\vartheta}{u_\tau} = 3.15 \times 10^{-5} m$$



Now, target $y^+ = 45$ for $k - \epsilon$ solver, as standard k-ε with wall functions requires the first cell center to sit in the log-law region: 30 to 300.

$$y_1 = y^+ . \delta_\vartheta \approx 1.45mm$$

This is the distance from the wall to the first cell centre — in blockMesh this means the first cell height is 2 × 1.45 mm ≈ 2.9 mm

k-ϵ solver for Channel Flow

- Since our first cell length should be $\approx 3mm$, and the height is 100mm, Therefore we can keep only 30-35 cells in our mesh in order to achieve the desired y+.

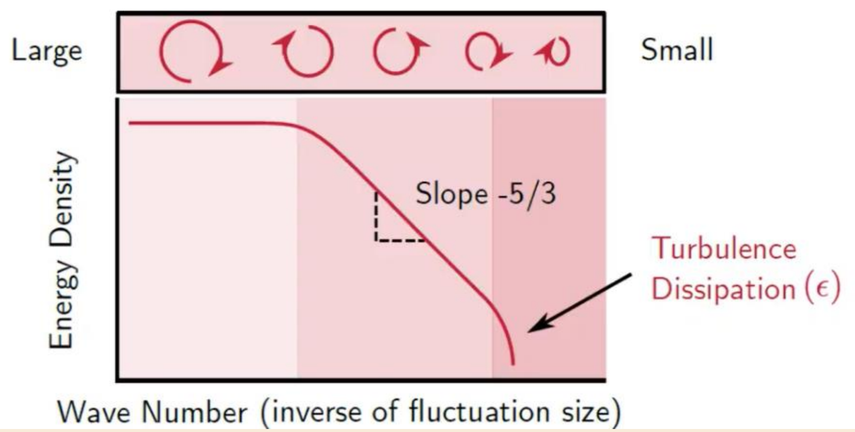
- Now for pipe problem, we can assume turbulence intensity to be 4% as a safe value.

$$k = \frac{3}{2} I^2 |U|^2 = 0.24\ m^2/s^2$$

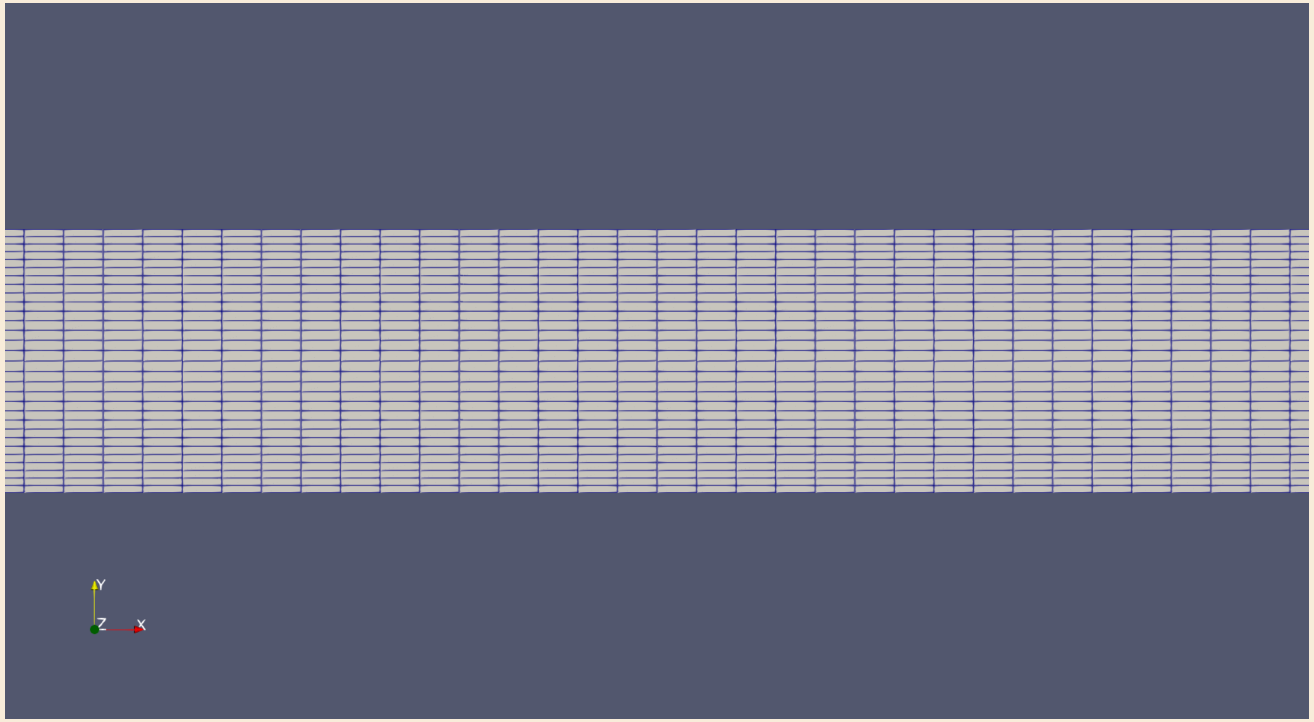
- For channel flow, the inlet mixing length is typically 7% of the total height. The inlet turbulence dissipation rate (ε) is estimated using a mixing-length model with a characteristic turbulence length scale of 0.007 m, giving an initial ε value of 2.76.

$$\epsilon = \frac{C_\mu^{\frac{3}{4}} k^{\frac{3}{2}}}{l} = 2.76\ m^2/s^3$$

ϵ is the turbulence dissipation rate. It is the rate that turbulent kinetic energy is converted into thermal energy by viscosity.



Mesh Details



Domain	12m × 0.1m × 0.1m
Mesh	800×30×1 = 24,000 cells
Solver	simpleFoam
Turbulence	k-epsilon standard
Inlet U	10 m/s
Reynolds number	66,667
y+ average	42.72
$\vartheta: kinematic\ visocosity$	1.5×10^{-5}

Velocity profiles

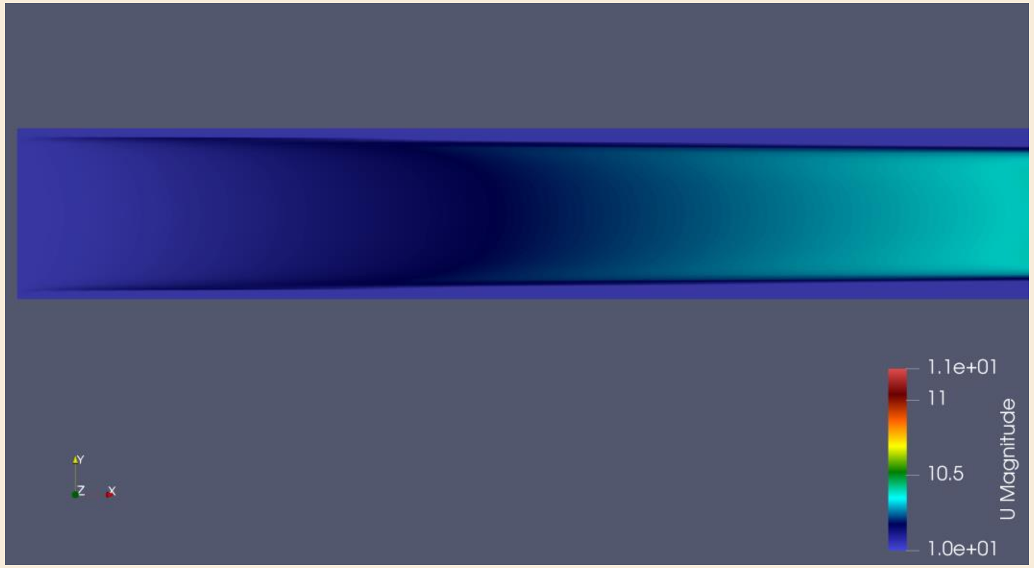


Fig. Developing Velocity profile near the inlet

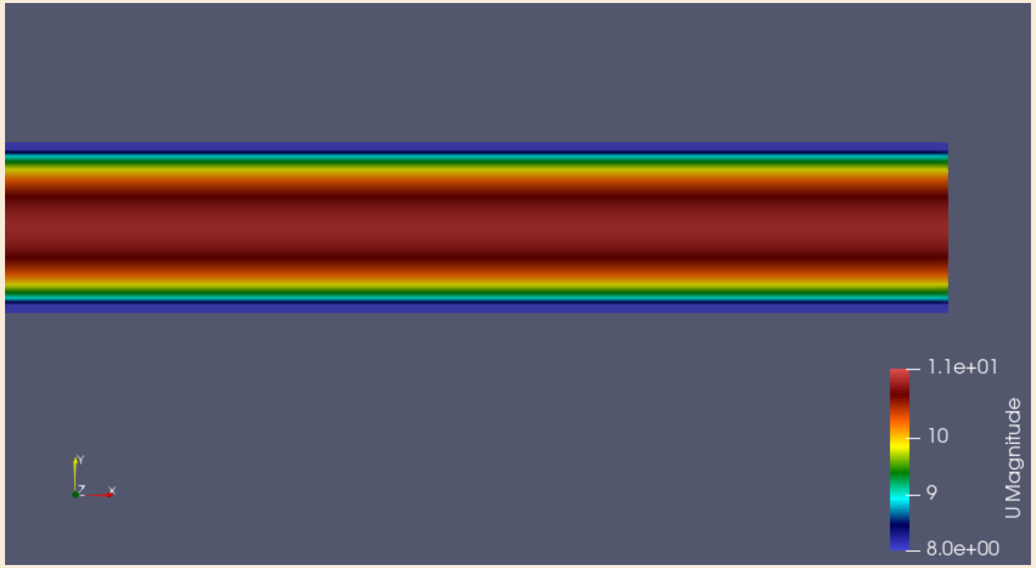


Fig. Developed Velocity profile near the outlet

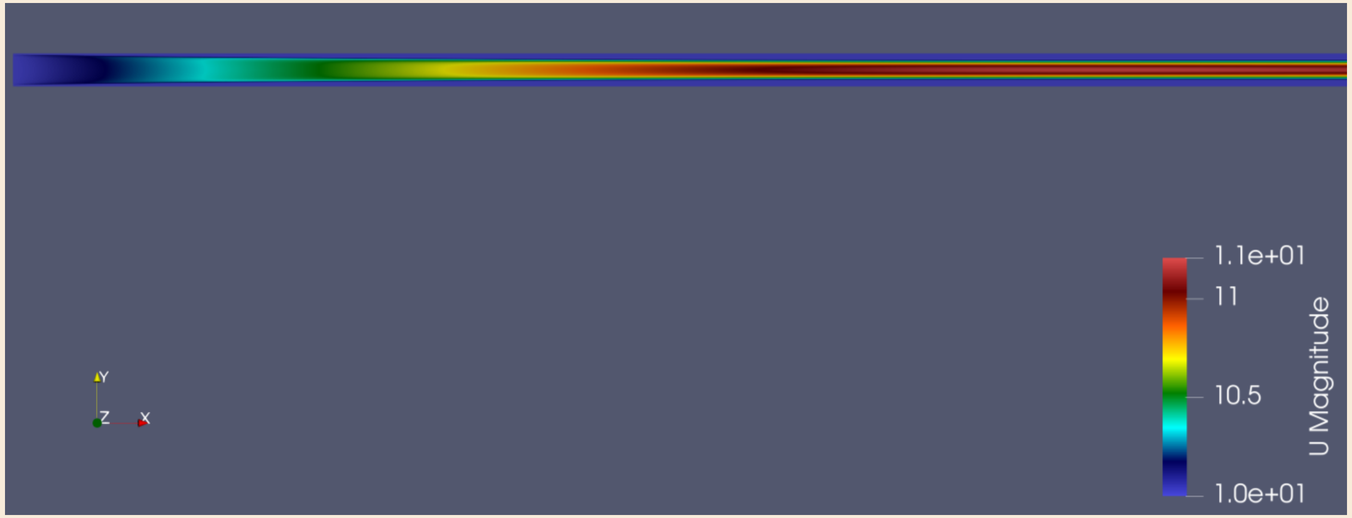
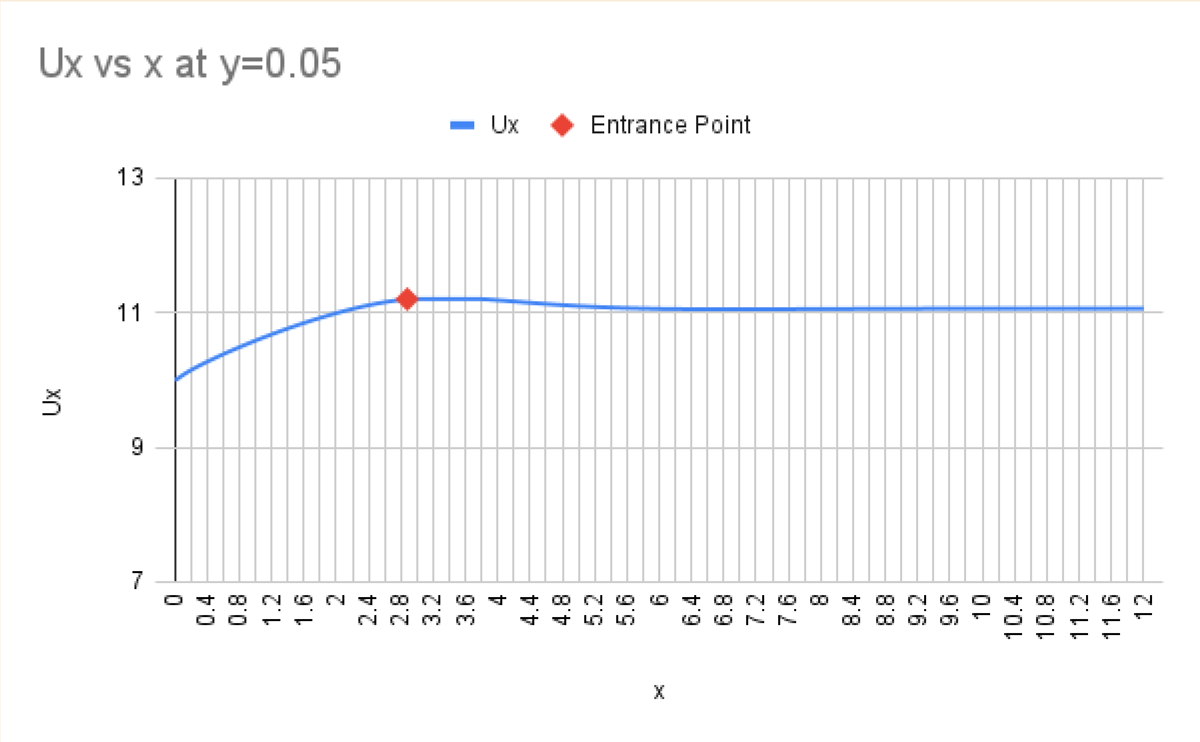
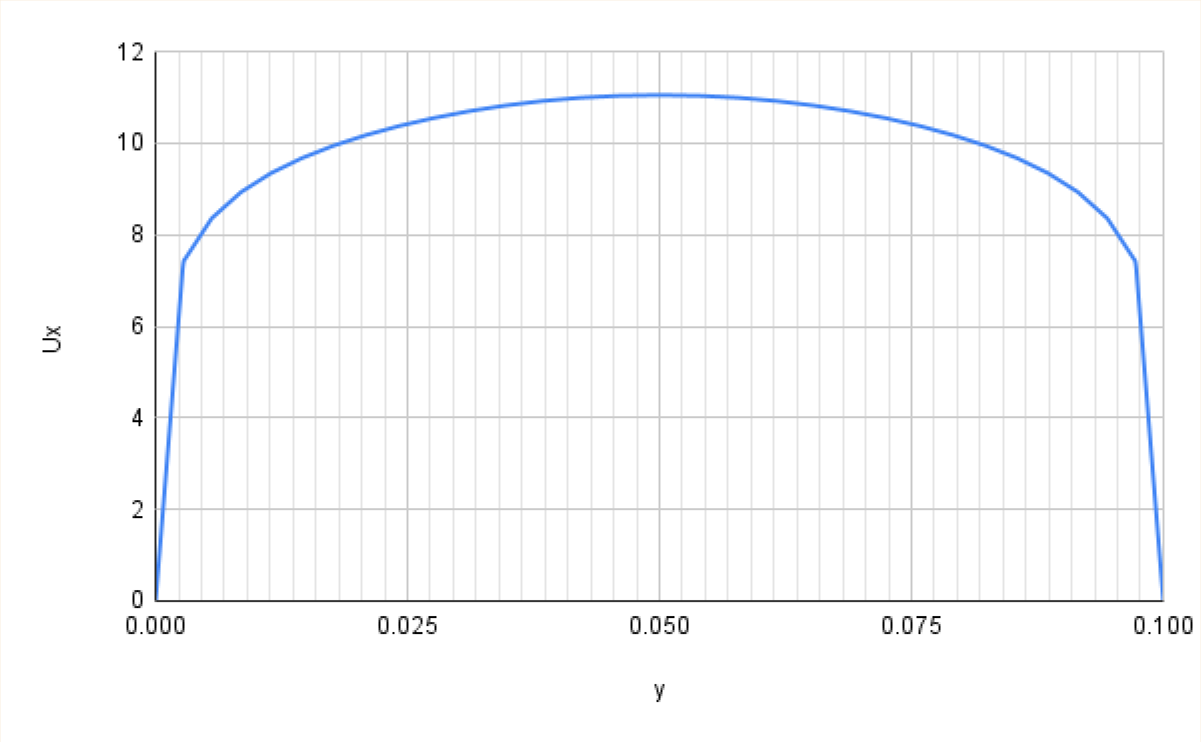


Fig. Overall Velocity profile

Velocity Distribution Comparison



Ux vs x at y = 0.05m



Ux vs y at x = 6m

- Before x=2.88m, the velocity profile is still developing as the boundary layers grow from both walls toward the centreline. Beyond x=2.88m, the boundary layers merge and the velocity profile no longer changes along the channel length, indicating fully developed turbulent flow.

In turbulent flow the boundary layers grow faster, and L_e is relatively shorter, according to the approximation for smooth walls

$$\frac{L_e}{d} \approx 4.4\, \mathrm{Re}_d^{1/6} \qquad \text{turbulent} \qquad (6.6)$$

Theoretically,

$$L_{e\approx}4.4 \times Re_d^{\frac{1}{6}} = 2.801$$

OpenFOAM Result:

$$L_e = 2.88$$

Pressure plot

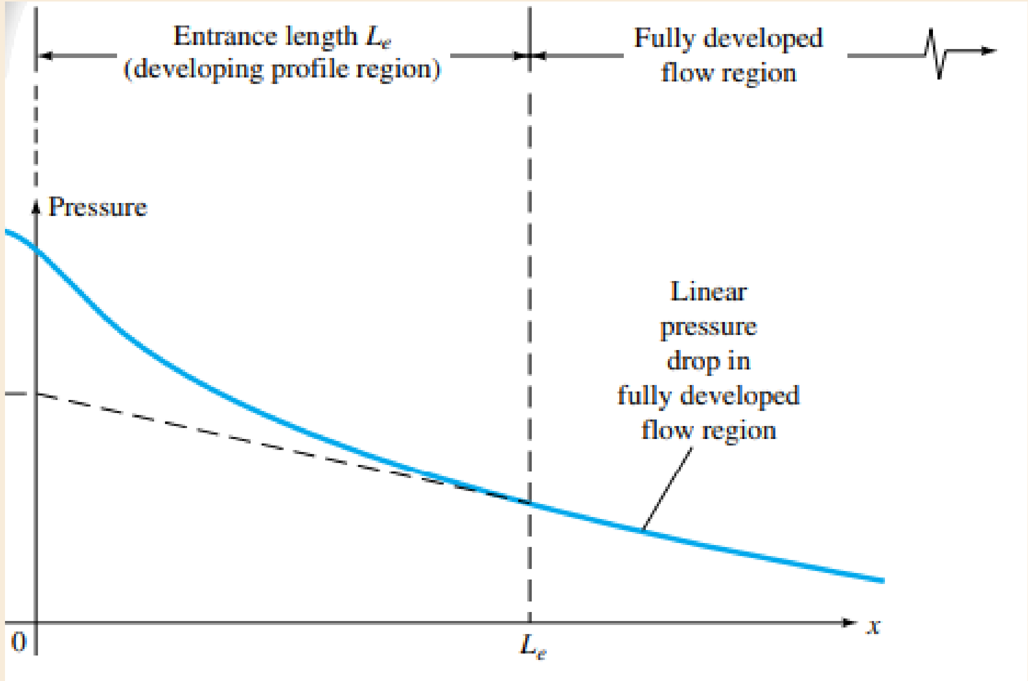


Fig. Pressure changes in the entrance of a duct flow.

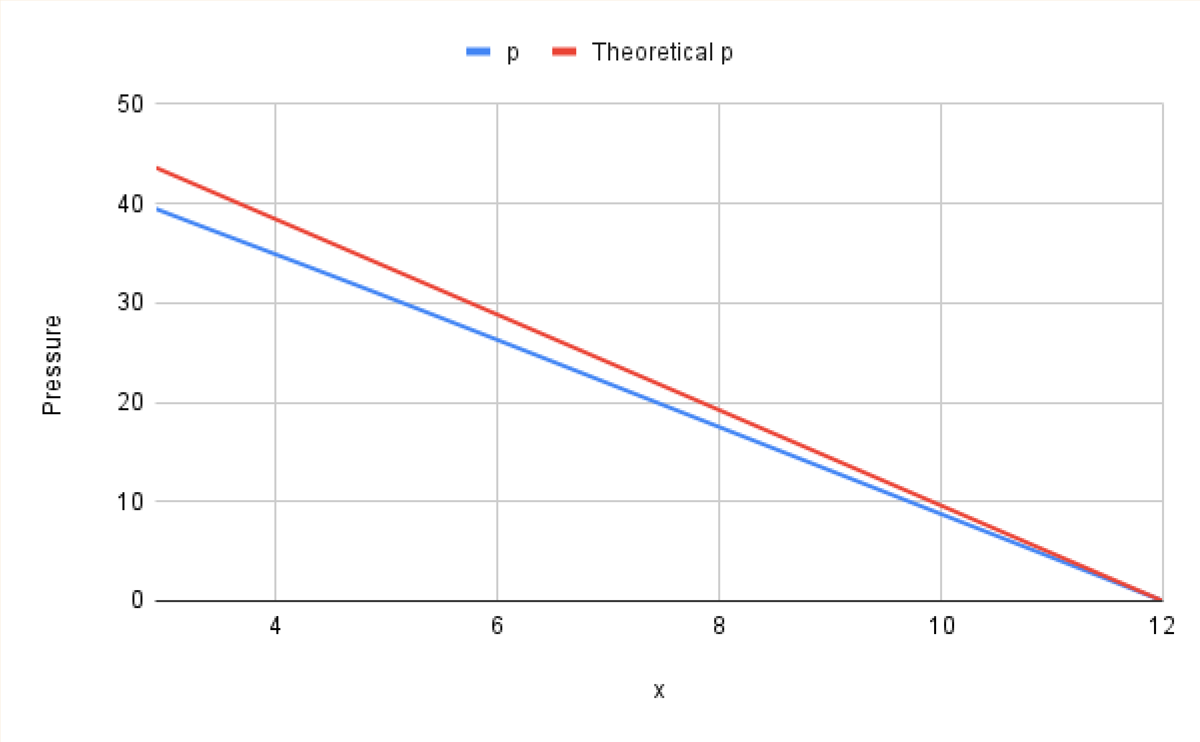


Fig. Pressure comparison- OpenFOAM values vs Theoretical Values

- The simulated pressure drops linearly along the channel length, consistent with the theoretical behaviour of fully developed duct flow where viscous losses dominate.

$$\frac{\partial p}{\partial x} = -\frac{2\tau_w}{H}$$

Theoretical Pressure gradient = -4.8
Graphical Result = -4.35

Law of the Wall

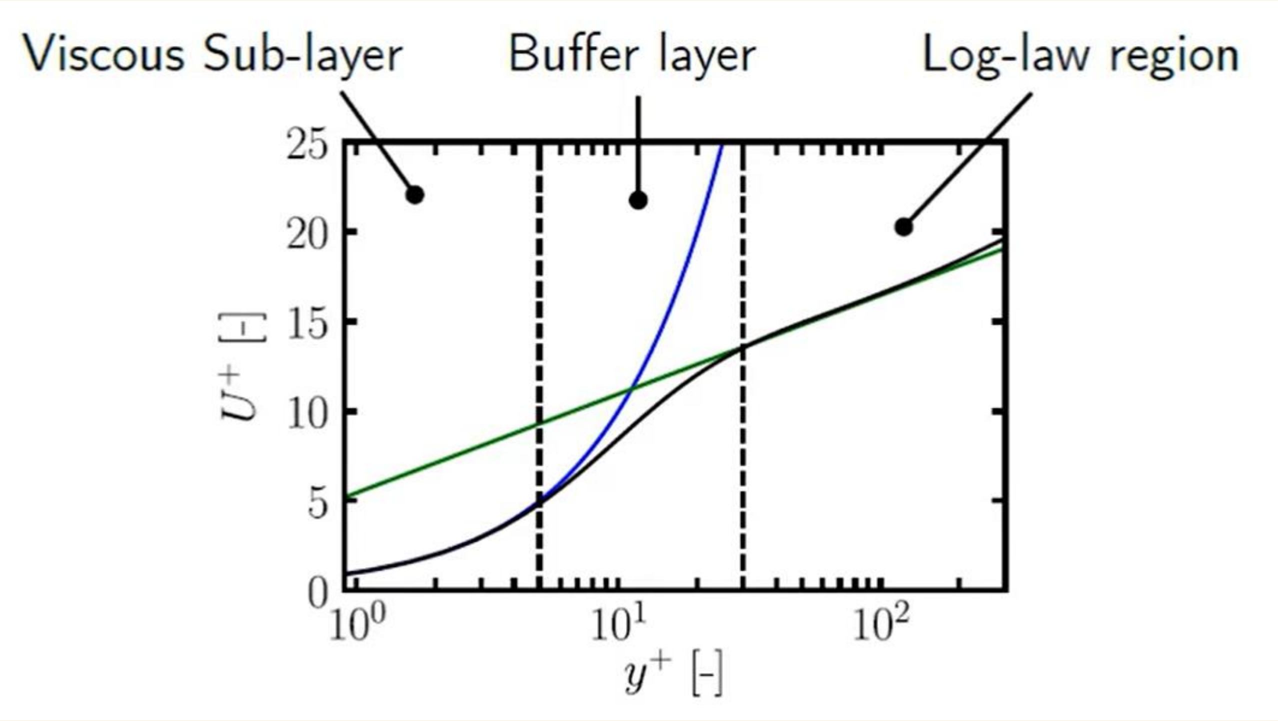
- The plot shows the dimensionless velocity U + U + vs. dimensionless wall distance y + y + , which describes how fluid velocity changes very close to a wall in turbulent flow.

- Three regions exist near the wall:

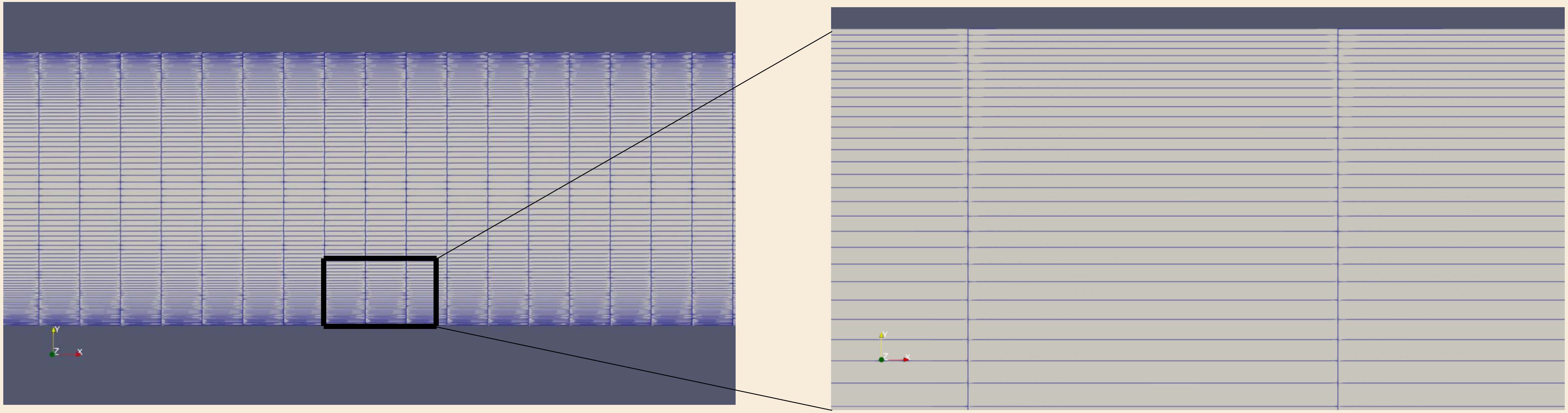
1. Viscous sublayer (y + < 5 y + <5): velocity increases linearly with distance from the wall.
2. Buffer layer (5 < y + < 30 5<y + <30): transition region where both viscous and turbulent effects are important.
3. Log-law region (y + > 30 y + >30): velocity follows a logarithmic relationship with distance.

- The black curve represents the actual turbulent velocity profile, while the straight and curved lines show the theoretical viscous and logarithmic laws.

- Turbulence models using wall functions (like k–ε) require the first mesh cell to lie in the log-law region (y + ≈ 30 – 100 y + ≈30–100) for accurate wall modeling.



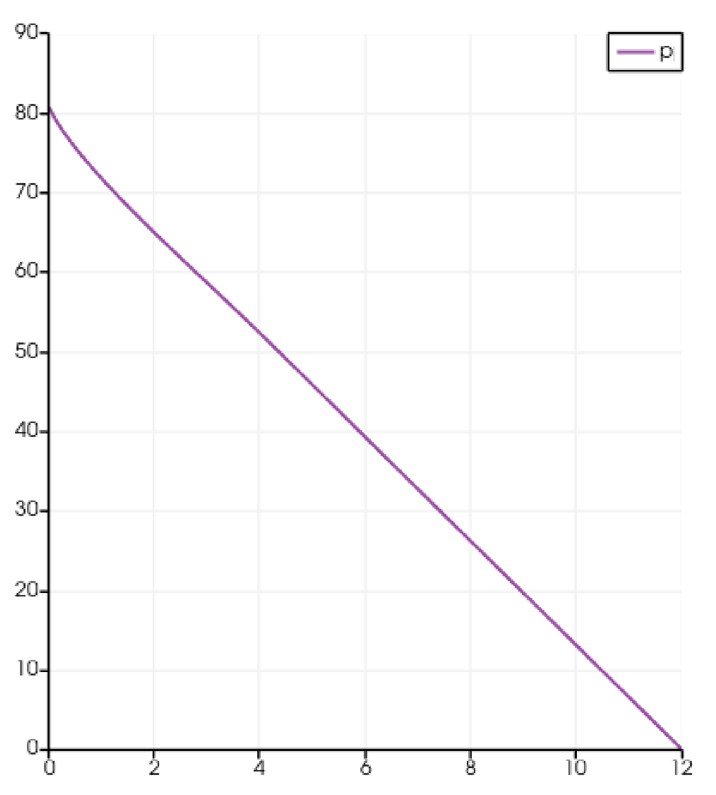
Attempt with a Finer Mesh



Mesh stats	Value
points:	161802
faces:	320900
internal faces:	159100
cells:	80000
faces per cell:	6
boundary patches:	4

- Used simple grading (1 10 1) for the upper half, and (1 0.1 1) for the lower half, to implement mesh grading near the walls
- Average y+ = 4.84 : Viscous Sub-Layer

New Results



Theoretical Pressure gradient = -4.8

Graphical Result = -6.6

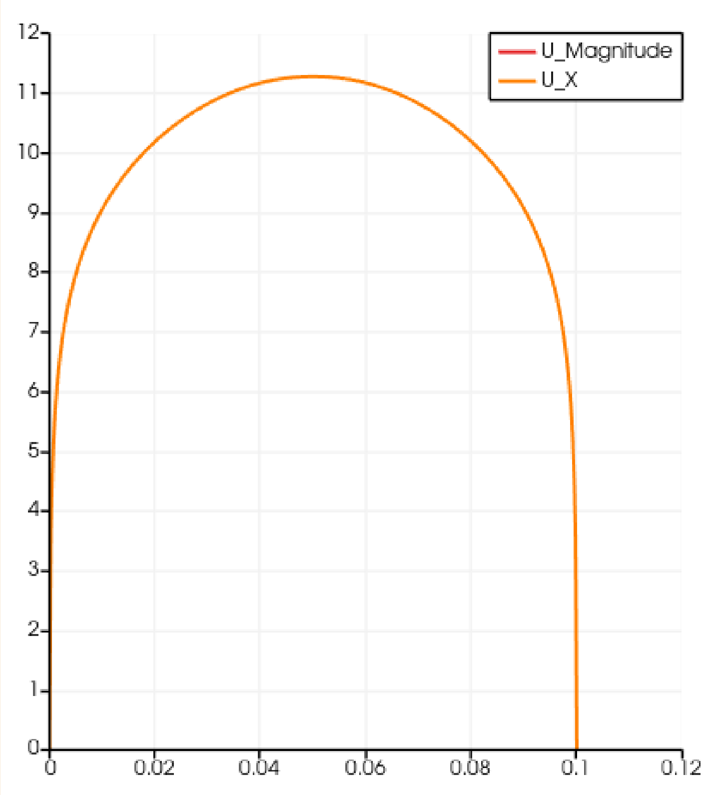


Fig. Velocity variation along y at x =6m

- The pressure gradient variation arises from mesh resolution and assumptions in the standard k–ε wall function model.
- The k–ε wall function assumes the first cell lies in the log-law region and applies the logarithmic velocity relation:

$$U^+ = \frac{1}{\kappa}\ln(y^+) + B$$

- In this case, $y^+ = 4.84$places the first cell in the viscous sublayer, where the correct relation is:

$$u^+ = y^+$$

- Applying the log-law in the viscous region leads to an incorrect wall velocity gradient, causing errors in wall shear stress and the resulting pressure gradient across the channel.

- Thus, even with a fine mesh, the standard k–ε turbulence model remains unsuitable for channel flow simulations because its wall-function formulation fails to accurately represent the near-wall viscous region.

$k - \omega$ Turbulence Model

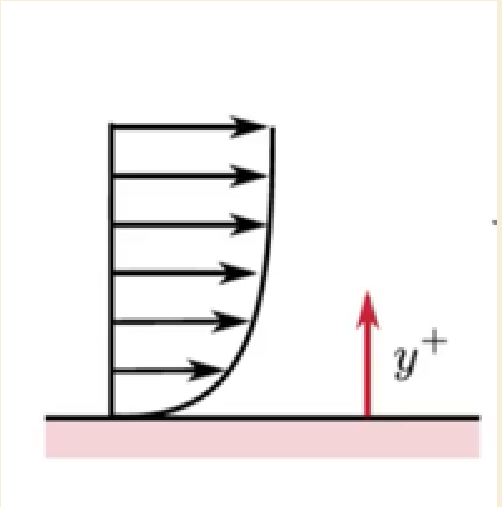
- ω is the specific turbulence dissipation rate (turbulence frequency). It is :

$$\omega = \frac{\epsilon}{C_\mu k} ; C_\mu = 0.09$$

- Transport Equation for ω :

$$\frac{d\rho\omega}{dt} + \nabla.(\rho U\omega) = \nabla.[(\mu + \mu_t)(\nabla\omega)] + \frac{\gamma}{v_t} P_k - \beta\rho\omega^2$$

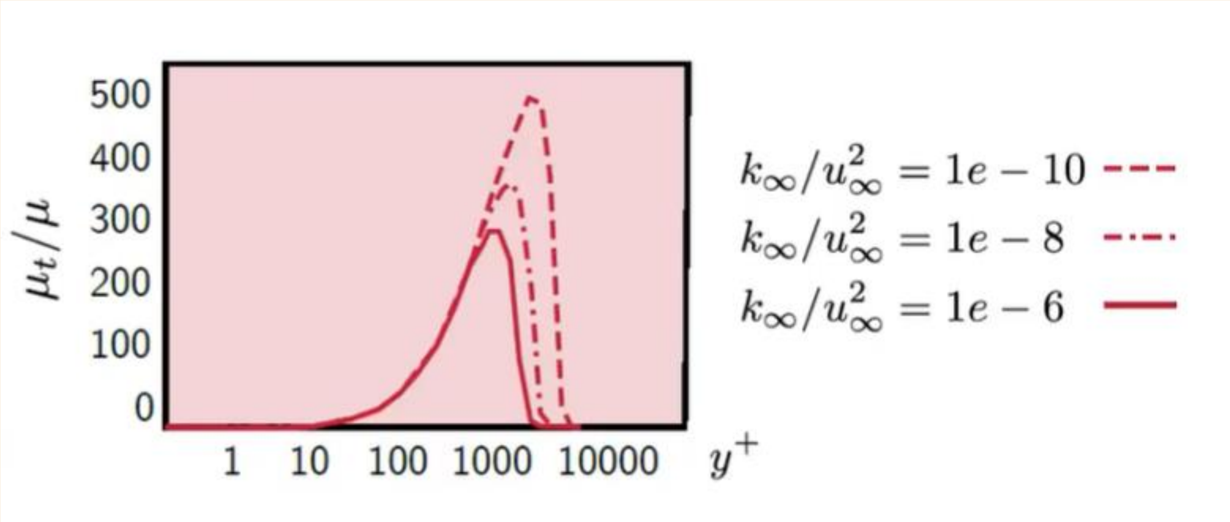
- Weakness of $k - \omega$ model: dependant on the freestream turbulence conditions. (i.e small changes in initial turbulence conditions can lead to major difference in results.)



- This is due to the missing cross diffusion term:

$$\frac{2\rho\sigma_{\omega 2}}{\omega} \cdot \frac{\partial k}{\partial x_j} \cdot \frac{\partial \omega}{\partial x_j}$$

- **Thus, a blend of both $k - \omega$ and $k - \epsilon$ is preferred.**

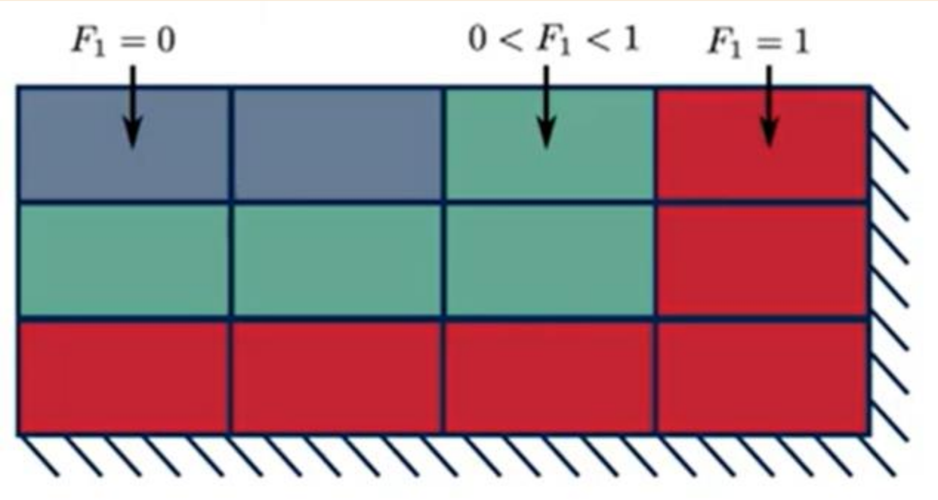


$k - \omega$ SST Turbulence Model

- The difference between $k - \epsilon$ and $k - \omega$ model is the presence of cross diffusion term in epsilon model while missing in the omega model.
- Therefore, we introduce a blending Function F_1 :

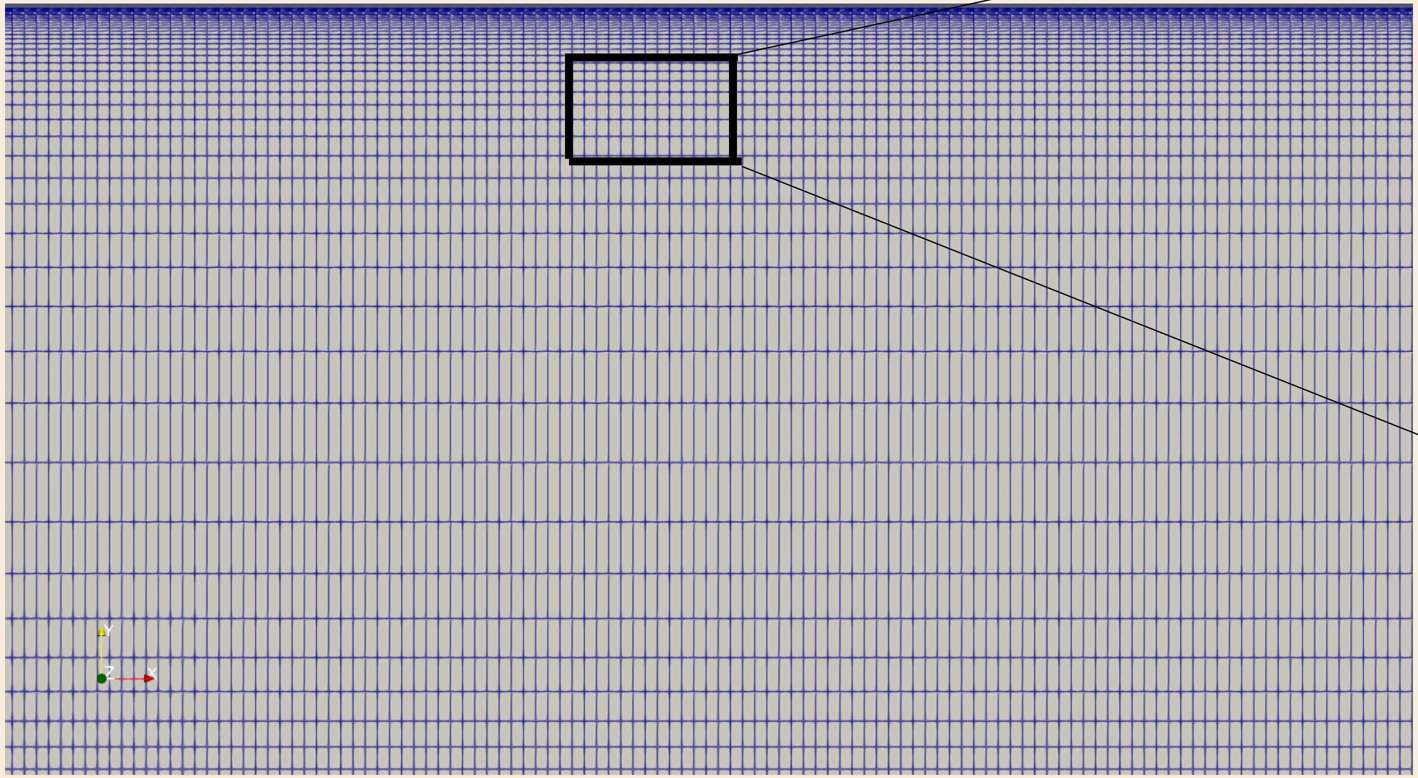
$$\frac{2(1-F_1)\rho\sigma_{\omega 2}}{\omega} \cdot \frac{\partial k}{\partial x_j} \cdot \frac{\partial \omega}{\partial x_j}$$

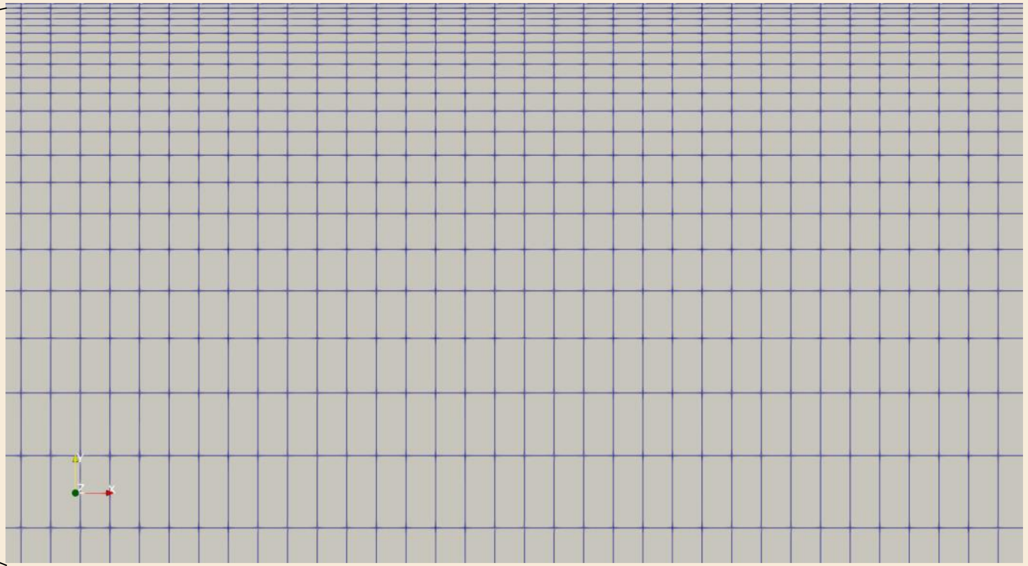
- $F_1 = 0 : k - \epsilon$ model
- $F_1 = 1 : k - \omega$ model
- This gives us the $k - \omega$ BST Turbulence Model, but it overpredicts the wall shear stress when compared to empirical data.
- <u>SST model uses viscosity limiter.</u>
- The limiter function F2 results in better agreement with experimental measures of separated flow.



Channel Flow : $k - \omega$ SST model

MESH DETAILS





As derived before,

$$Viscous\ Length\ scale\colon\ \delta_{\vartheta} = \frac{\vartheta}{u_{\tau}} = 3.15 \times 10^{-5} m$$

Targeting a y+ value close to 1, first cell will have a height of 0.063 mm .
The mesh contains 92 cells in the y direction, with heavy grading near the wall(ratio of 538).

Mesh stats:

points: 1674186
faces: 3321092
internal faces: 1646908
cells: 828000

Velocity Contours

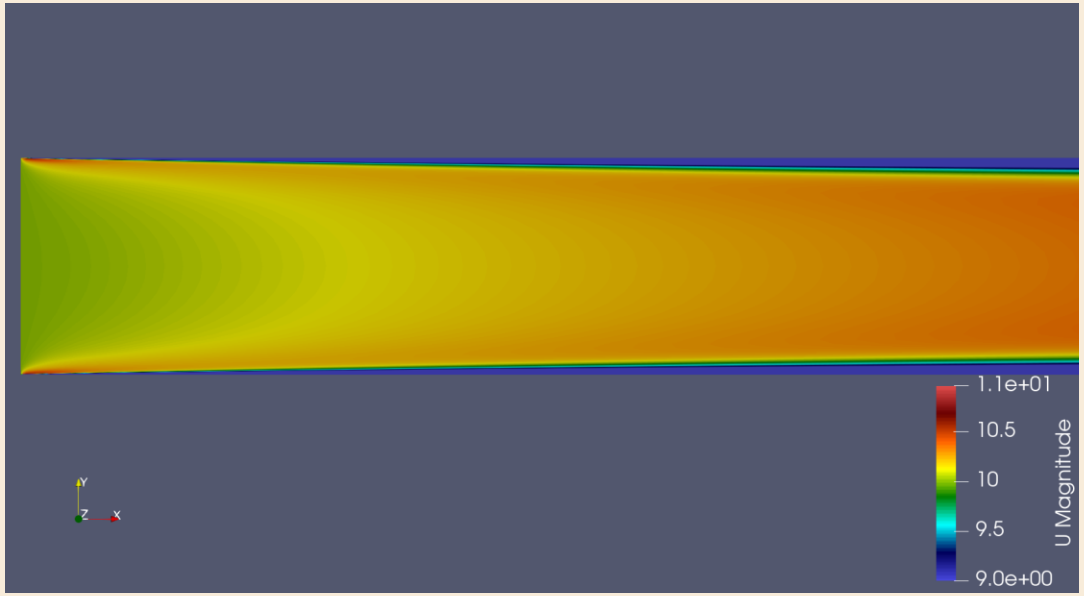


Fig. Developing Velocity profile near the inlet

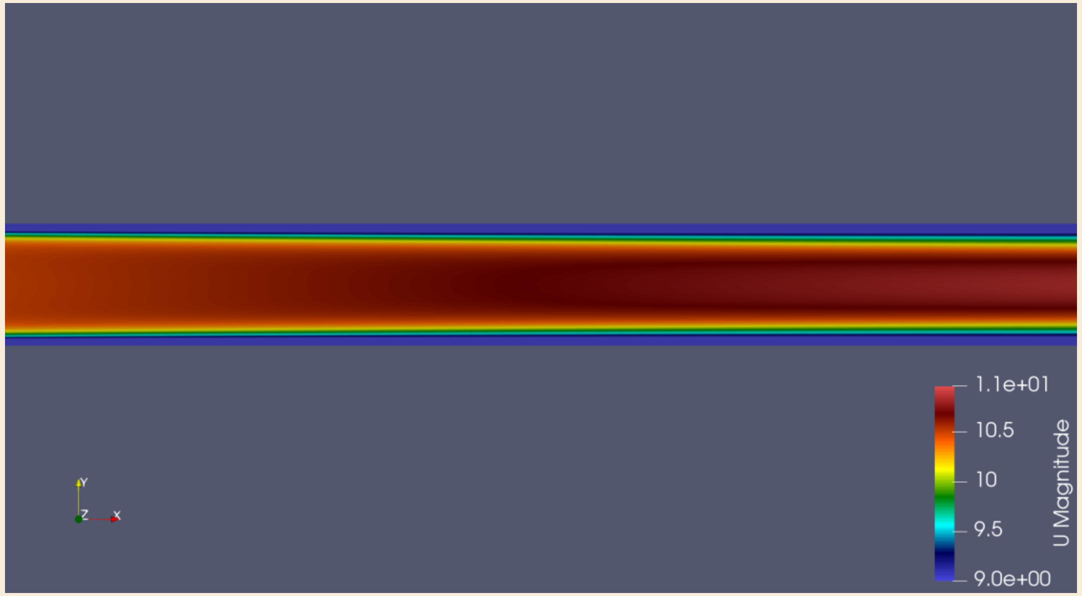


Fig. Developed Velocity profile near the outlet

Fig. Overall Velocity profile

Velocity and Pressure Plots

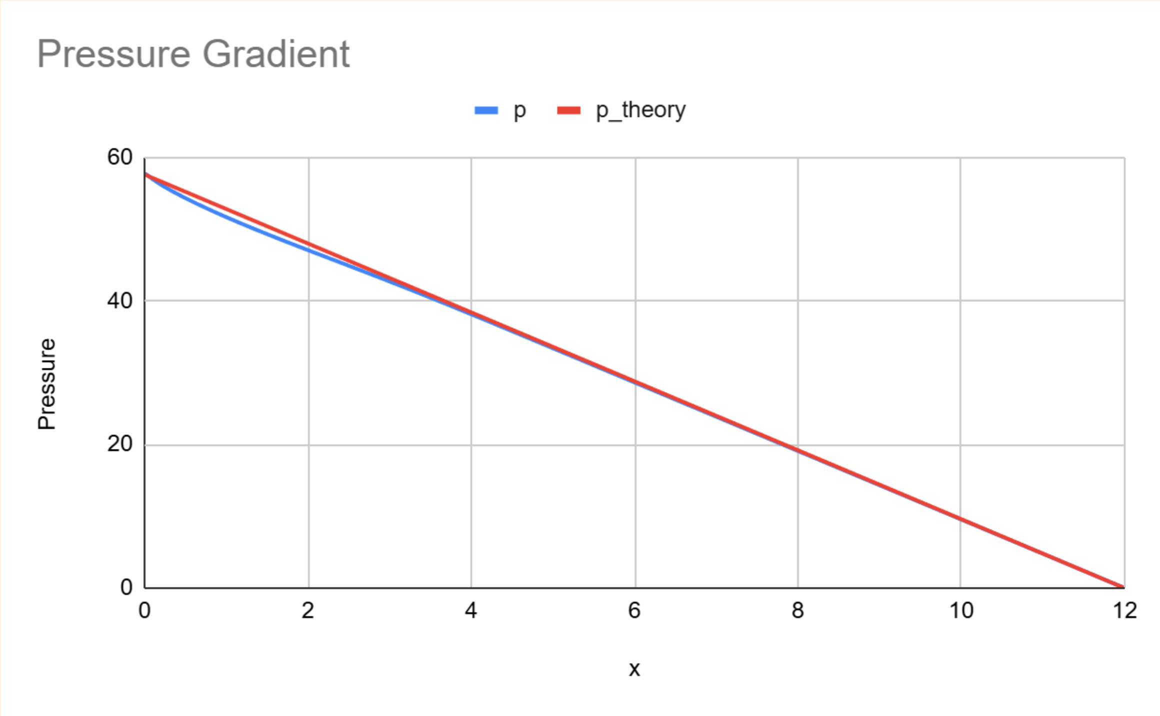


Fig. Pressure comparison- OpenFOAM values vs Theoretical Values

$$\frac{\partial p}{\partial x} = -\frac{2\tau_w}{H}$$

Theoretical Pressure gradient = -4.8

Graphical Result = -4.73

Difference: 1.45%

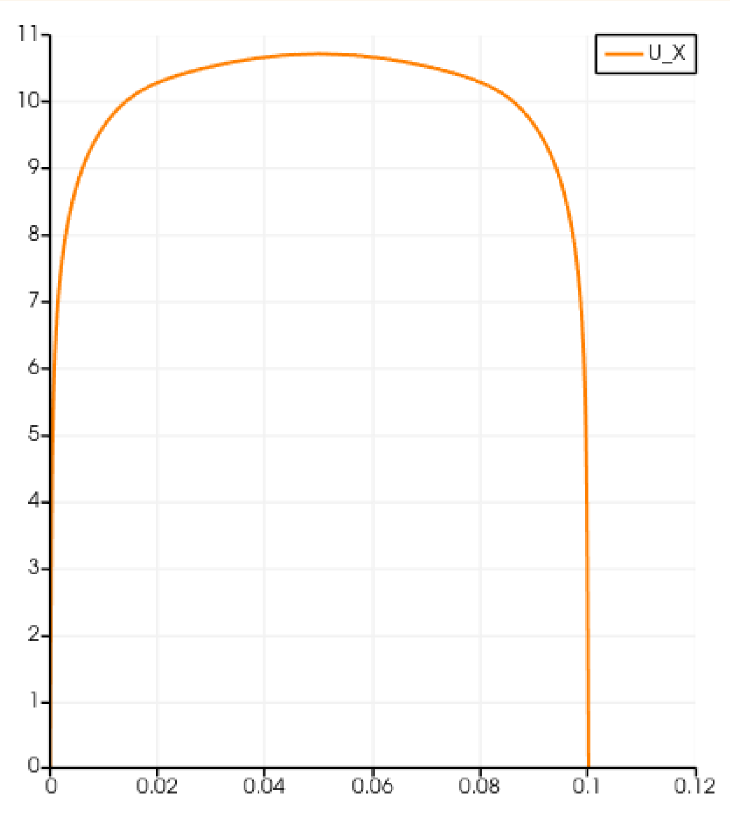


Fig. Velocity variation along y at x =10m

Tke contour and plot

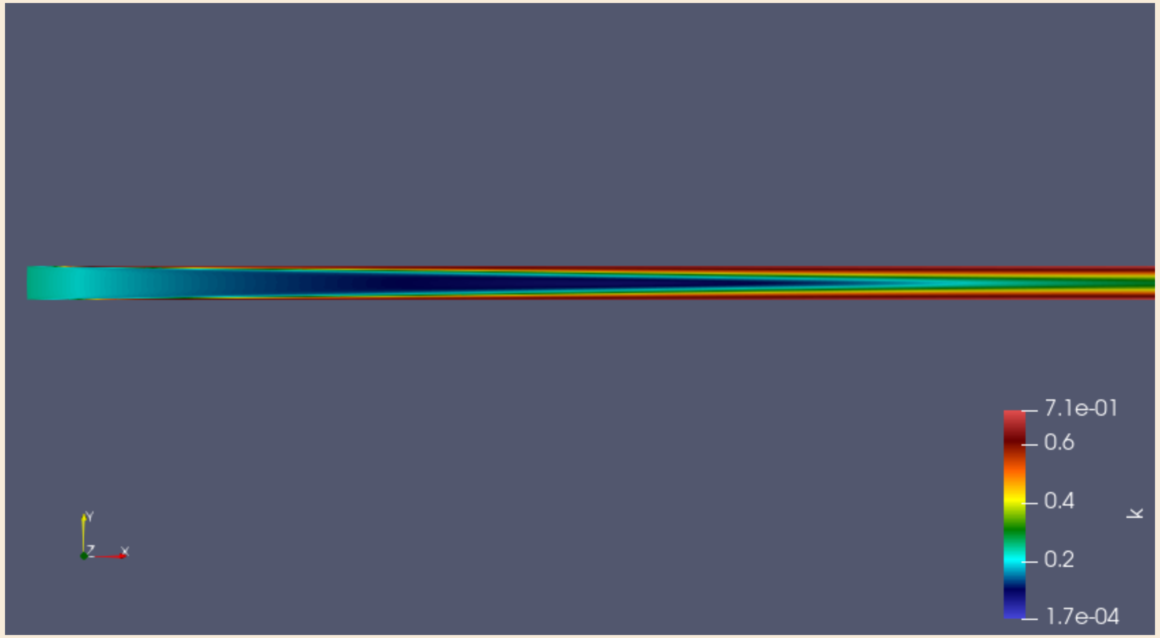


Fig. turbulent kinetic energy contour plot

- Turbulent kinetic energy is near-zero at the inlet and develops along the streamwise direction, with high-k regions (red) growing from both walls where velocity gradients and shear production are largest
- The low-k core (blue) at the channel center persists throughout the domain, consistent with near-zero velocity gradient at the centerline and hence negligible local turbulence production
- Beyond x ≈ 3m the near-wall k bands become uniform and unchanging, indicating the flow has reached a **fully developed turbulent state**

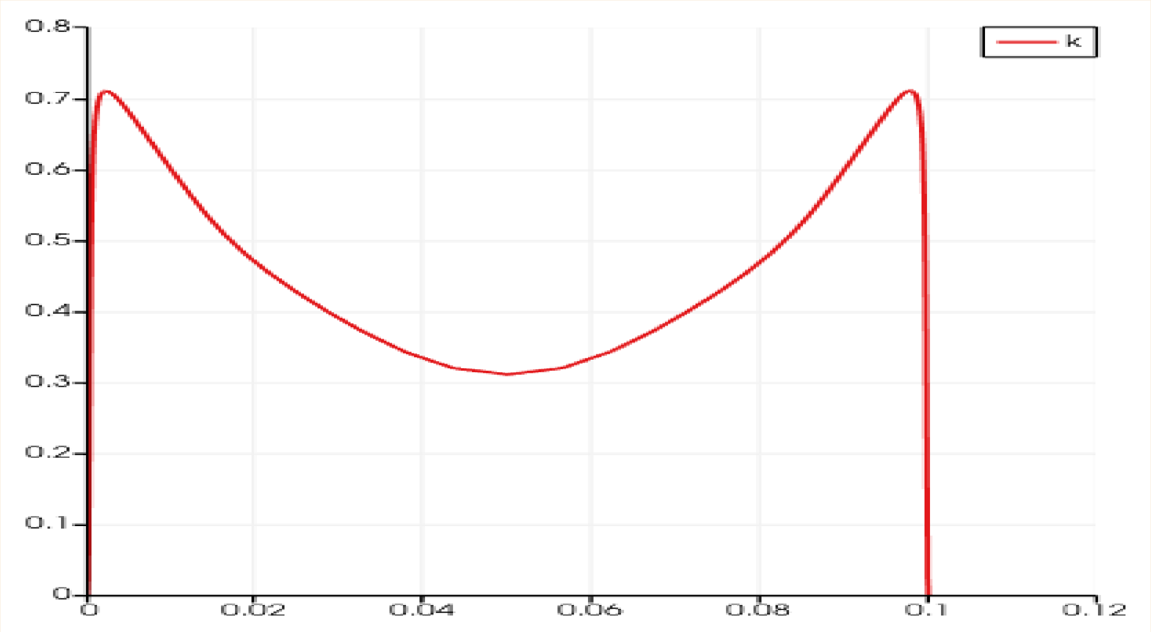


Fig. k vs y at x=10m(Developed flow)

- Peak turbulent kinetic energy of ~0.70 m²/s² occurs adjacent to both walls where shear production $P_k = \vartheta_t.(\frac{\partial U}{\partial y})^2$ is maximum, decaying to a minimum of ~0.32 m²/s² at the centerline
- At the wall itself k drops to zero, consistent with the no-slip condition which suppresses all velocity fluctuations at a solid boundary
- The non-zero k at the centerline, despite zero local production, is sustained by **turbulent diffusion** transporting energy outward from the near-wall region

Other Plots

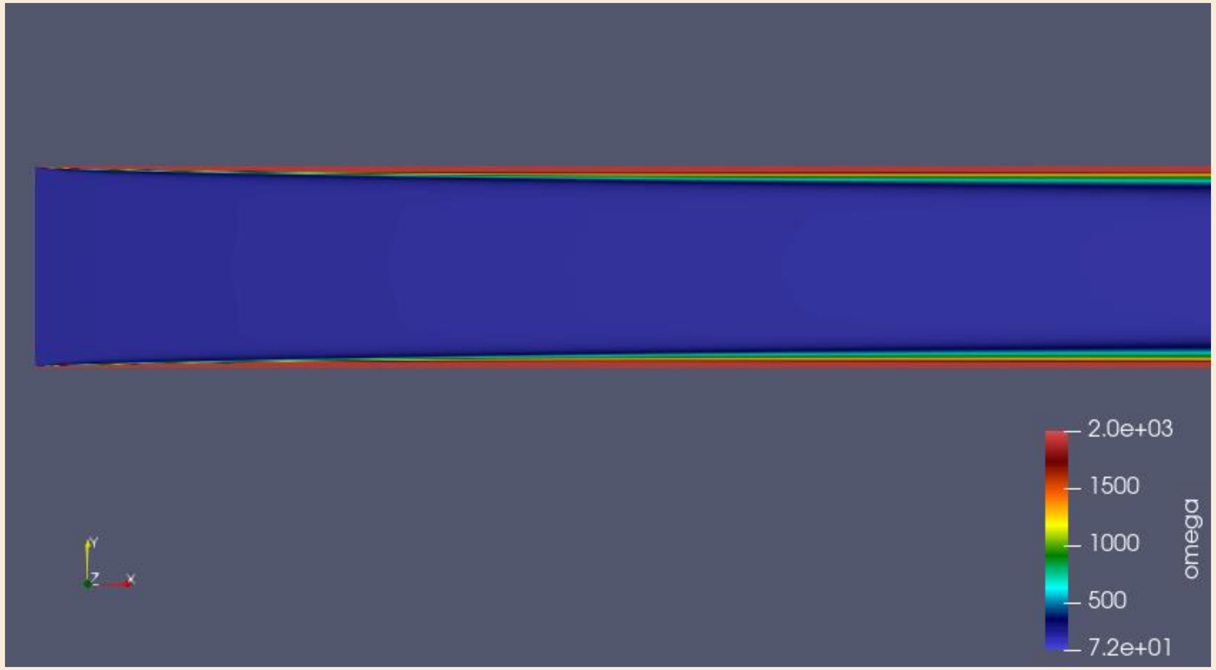


Fig. specific dissipation rate (omega) contour plot

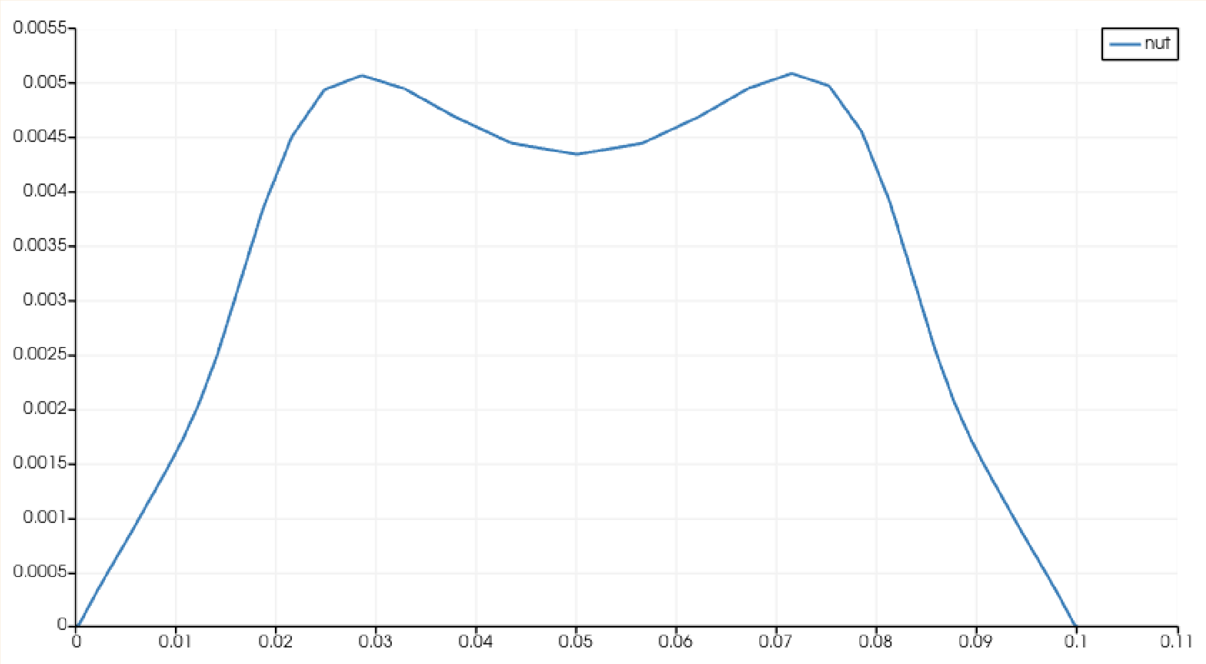


Fig. eddy viscosity vs y at x=10m

- The contour shows ω is extremely large in thin layer at both walls and near-uniform throughout the bulk, reflecting the **inverse dependence ω ~ 1/y** as the wall is approached — turbulent length scales shrink to near-zero as eddies are physically constrained by the solid boundary

- Away from the walls, turbulent energy is dissipated slowly over large length scales, which is why ω remains low and uniform across the entire channel core

- Eddy viscosity rises from zero at both walls, where turbulent mixing is suppressed by the no-slip condition, and peaks near y ≈ 0.025m and y ≈ 0.075m where the balance between turbulence production and dissipation is greatest

- The **F2 blending function** in SST activates near the wall and modifies eddy viscosity ϑ_t . Near the centerline where shear S is very low, the limiter switches and causes the slight central dip.

- This is to prevent unphysical over-production of eddy viscosity, a behaviour absent in standard k-ε

Law of the Wall Validation — k-ω SST Channel Flow

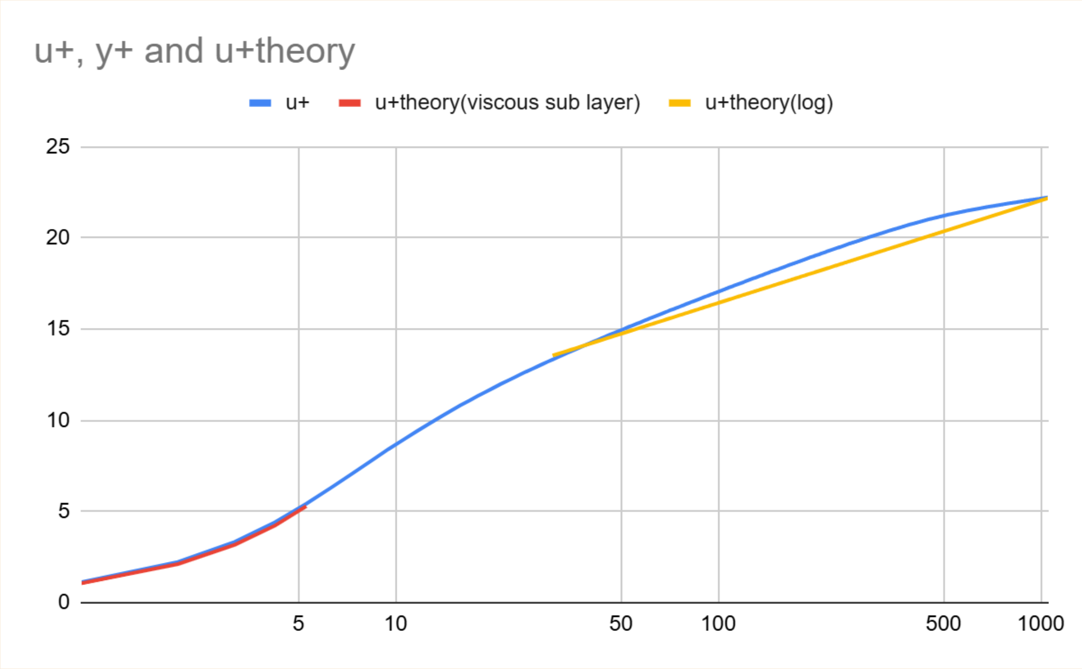
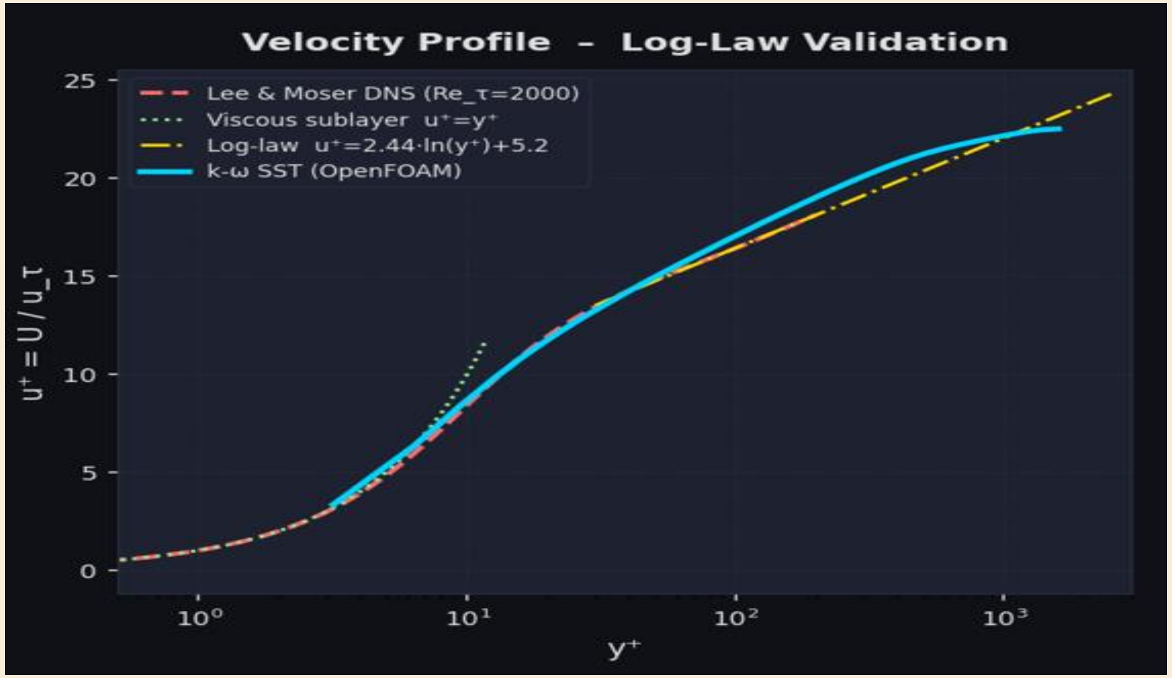


Fig. u+ vs y+

- In the viscous sublayer ($y^+ < 5$) the simulated velocity profile follows $u^+ = y^+$ exactly, confirming that the near-wall mesh resolution is sufficient to fully resolve the viscous-dominated region without wall functions

- In the log-law region ($y^+ > 30$) the simulated profile closely follows $u^+ = (1/\kappa)\cdot\ln(y^+) + 5.2$, confirming fully developed turbulent behaviour and **validating the k-ω SST model** against well-established theory



- velocity profile validation plot comparing k-ω SST OpenFOAM results against the viscous sublayer, log-law, and Lee & Moser DNS data across all three near-wall regions in wall units. (AI generated)

https://www.researchgate.net/profile/John-Kim-23/publication/243777258_Direct_Numerical_Simulation_of_Turbulent_Channel_Flow_up_to_Re590/links/5691fdfa08aec14fa55b9c52/Direct-Numerical-Simulation-of-Turbulent-Channel-Flow-up-to-Re590.pdf

Next Target